

PAPER-FBT10A

Mass and Gravity as Dual Readout Sectors of Hamiltonian Structural Dynamics

From Thimble-Depth Inertia and Invariant-Order Coarse-Graining
 to the Constitutive Response Metric and Conditional Gravitational Closure

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Abstract

The present paper formulates FBT10A as the constitutive bridge between microscopic structural dynamics, thimble-depth inertia, the convergent probability architecture of the FBT06 sequence, invariant-constrained macroscopic coarse-graining, and the effective Lorentzian metric used by the downstream gravitational closure.

The global six-dimensional/four-dimensional organisation is inherited from FBT0B,

$$T_{\text{rel}}^2 \hookrightarrow X_{\text{Reg}}^{(6)} \xrightarrow{\pi_{\text{fib}}} B_{\text{Reg}}^{(4)}.$$

On the regular Lorentzian branch, the four-dimensional base carries the signature and causal structure supplied by FBT01E and FBT08A. FBT10A does not introduce a second competing projection from the carrier to spacetime. The bundle projection is distinguished from tensorial readout descent, probability assembly, first-order geometric readability, invariant-order readout quotienting, adiabatic low-energy selection, and final macroscopic resolution.

The paper also separates two upstream flows. Hamiltonian structural evolution is generated by

$$\iota_{X_{H_{\text{str}}}} \Omega_{\mathcal{B}} = dH_{\text{str}},$$

whereas Lefschetz-thimble selection is governed by a complex Morse/gradients flow associated with the real part of the complex action. The two flows may be compatible on an admissible branch, but they are not identified. Hamiltonian evolution transports the physical structural state; the Morse flow selects and organises the integration cycles used in the semiclassical sector decomposition.

At the microscopic mechanical level, an admissible thimble sector carries a lifted suppression depth $\hat{D}_i$. The mass-depth relation is treated as a constitutive leading-order hypothesis,

$$M_i = M_* Z_i^{\text{pref}} e^{-\hat{D}_i},$$

with Z_i^{pref} collecting admissible prefactor, normalisation, and coherence contributions. Under the collective-coordinate construction of FBT08E, the reduced stability architecture of FBT15C and FBT08F, and the dynamical adiabatic condition

$$\varepsilon_{\text{ad}} := \frac{\Omega_{\text{ext}}}{\omega_{\text{gap}}} \ll 1,$$

the leading reparametrisation-invariant worldline term is

$$S_{\text{eff}}^{(i)} = -M_i \int ds_{\text{eff}} + S_{\text{corr}}^{(i)}.$$

The probabilistic layer is inherited rather than postulated. FBT06C establishes absolute convergence on the unique admissible fibrewise thimble, FBT06D constructs the normalised fibre probability measures ν_b , and FBT06E assembles those measures over the regular relative-phase bundle into an admissible measurable structure μ_{adm} . FBT10A therefore introduces no second map from a complex thimble amplitude to a real probability.

The first macroscopic interface is geometric readability. The FBT01A mixed form

$$\eta \in H_A^* \wedge V^*$$

defines

$$\eta_V^b : V \longrightarrow H_A^*, \quad v \longmapsto \iota_v \eta.$$

On the regular Liouville-saturated branch, $\omega_H \wedge \eta^{\wedge 2} \neq 0$ forces $\text{rank } \eta_V^b = 2$, so every vertical direction is mixed-readable. Thus η supplies a first-order transmission channel rather than a lossy coarse-graining by itself.

FBT0C v1.5 supplies the next, logically distinct layer. For each response channel r , an order-dependent readout map

$$\mathcal{Q}^{(r)} : \mathcal{E}_{\text{read}} \longrightarrow \mathcal{J}^{(r)}$$

defines the equivalence relation

$$\xi_1 \sim_r \xi_2 \iff \mathcal{Q}^{(r)}(\xi_1) = \mathcal{Q}^{(r)}(\xi_2),$$

and hence the invariant-order quotient

$$\Pi_{\text{inv}}^{(r)} : \mathcal{E}_{\text{read}} \longrightarrow \mathcal{E}_{\text{inv}}^{(r)} := \mathcal{E}_{\text{read}} / \sim_r.$$

Whenever $\mathcal{Q}^{(r)}$ is non-injective, distinct microscopic representatives become indistinguishable in that response channel. This is loss of readout distinguishability or non-reconstructibility, not destruction of microscopic information. Hamiltonian preservation of $\Omega_{\mathcal{B}}^{\wedge k}$ is therefore compatible with a many-to-one higher-order readout.

The response quotient is channel dependent. A linear channel may have $\Pi_{\text{inv}}^{(r)} = \text{id}$, whereas a quadratic metric channel may identify additional microscopic states. In particular, the local curvature-quadratic response

$$Q_{\mu\nu}[\mathcal{K}] = \langle \mathcal{K}_{\mu\alpha}, \mathcal{K}_{\nu\beta} \rangle_V H^{\alpha\beta}$$

satisfies

$$Q_{\mu\nu}[-\mathcal{K}] = Q_{\mu\nu}[\mathcal{K}].$$

Thus the type-changing response itself may induce a nontrivial readout quotient.

FBT08F supplies the dynamical filter. Assuming compatibility between the channel equivalence and the spectral projection, the positive frequency gap induces an adiabatic selector

$$\mathbf{P}_{\text{ad}}^{(r)} : \mathcal{E}_{\text{inv}}^{(r)} \longrightarrow \mathcal{E}_{\text{low}}^{(r)}$$

that retains the collective or explicitly enlarged low-energy sector while suppressing the gapped transverse complement up to $O(\varepsilon_{\text{ad}})$ corrections. A residual resolution map $\Pi_{\text{res}}^{(r)}$ then specifies which surviving invariant classes are resolved macroscopically.

Accordingly, each response channel has the factorised map

$$\boxed{\Pi_{\text{macro}}^{(r)} = \Pi_{\text{res}}^{(r)} \circ \mathbf{P}_{\text{ad}}^{(r)} \circ \Pi_{\text{inv}}^{(r)} \circ \mathbf{P}_{\eta}.$$

For the selected metric-response channel the superscript is suppressed. The assembled measure induces

$$\boxed{\mu_{\text{macro}} = (\Pi_{\text{macro}})_* \mu_{\text{adm}} = (\Pi_{\text{res}})_* (\mathbf{P}_{\text{ad}})_* (\Pi_{\text{inv}})_* (\mathbf{P}_{\eta})_* \mu_{\text{adm}}.}$$

Macroscopic response variables are conditional expectations with respect to this measure. In a sector-diagonal regime,

$$\langle \mathcal{O} \rangle_{\text{macro}} = \sum_i p_i \mathcal{O}_i, \quad p_i \geq 0, \quad \sum_i p_i = 1,$$

where p_i is the conditional measure of sector i , not an independently postulated physical weight. The dimensionless sector probability remains distinct from the dimensionful mass coefficient,

$$\boxed{M_i \neq p_i.}$$

Moreover, readout equivalence is channel dependent: two microscopic states can be indistinguishable to the metric channel while remaining distinguishable to the mass readout. If inter-sector coherence survives all channel quotients and low-energy/resolution filters, off-diagonal data ρ_{ij}^{ov} enter through an FBT27-compatible overlap construction.

The resulting constrained response package contains real, gauge-covariant macroscopic images of $\Omega_{\text{phase}}^{\text{eff}}$, the mixed form η , relative-phase connection curvature, and the Hermitian and antisymmetric parts of the readout QGT, together with μ_{macro} , any surviving ρ^{ov} , and the constitutive data $\mathcal{C}_{\text{macro}}$. A branch-dependent map

$$\mathcal{R}_{\text{met}} : \mathfrak{X}_{\text{macro}}^{\text{adm}} \longrightarrow \Gamma(\text{Sym}^2 T^* B_{\text{Reg}}^{(4)})$$

then gives

$$\boxed{g^{\text{eff}} = g^{(0)} + \mathcal{R}_{\text{met}}(\mathfrak{R}_{\text{macro}}).}$$

The genuinely residual freedom is concentrated primarily in the final channel resolution and the type-changing response-metric map. First- order readability, invariant-order indistinguishability, and adiabatic persistence are constrained upstream by FBT01A, FBT0C, and FBT08F.

A stable configuration can be described either as a variational worldline source or as a perturbation of the constitutive response data. These are candidate macroscopic descriptions subject to a downstream matching condition; they are not independent duplicate sources. The balanced-shell and level-24 logic is inherited unchanged from FBT01D and FBT28B. Finally, a conditional Palatini completion is restricted to a torsion-suppressed branch unless the resolved FBT08G spin current is retained, in which case an Einstein–Cartan-type extension may be required.

FBT10A therefore identifies mechanical mass, probability, and metric response as distinct readout channels on one microscopic structural space. Their unity is not an identity of masses, probabilities, invariants, or metrics. It is a controlled hierarchy of Hamiltonian transport, complex-Morse selection, convergence and measure assembly, first-order transmission, invariant-order quotienting, adiabatic low-energy selection, residual resolution, constitutive metric response, and conditional gravitational closure.

Paper-specific keywords: Hamiltonian structural dynamics; Lefschetz–Morse selection; thimble suppression depth; inertial mass; worldline effective action; mixed-geometric readability; invariant-order readout quotient; adiabatic low-energy selection; channel-dependent macroscopic coarse-graining; response-metric map; conditional expectation; inter-sector overlap; constrained response data; conditional Palatini closure.

Framework keywords: Fracture–Berry–Tension framework; FBT0B; FBT0C; FBT01A; FBT01D; FBT01E; FBT06C; FBT06D; FBT06E; FBT08A; FBT08E; FBT08F; FBT08G; FBT12A; FBT12C; FBT12D; FBT15C; FBT27 sequence; quantized phase flux; affine level; readout versus observation.

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1 Introduction

The mature FBT hierarchy separates structures that earlier versions of the programme tended to compress into one geometric statement. The present paper is organised around the following distinctions:

bundle projection $\neq$ tensorial readout descent,
 Hamiltonian physical flow $\neq$ Lefschetz–Morse selection flow,
 phase curvature $\neq$ response metric $\neq$ Riemann curvature,
 static Hessian gap $\neq$ dynamical frequency gap,
 first-order readability $\neq$ invariant-order distinguishability,
 invariant-order distinguishability $\neq$ adiabatic persistence $\neq$ macroscopic resolution,
 fibre probability normalisation $\neq$ macroscopic coarse-graining,
 diagonal sector probability $\neq$ inter-sector overlap/coherence,
 constitutive metric response $\neq$ an independently duplicated matter source.

The global six-dimensional organisation is fixed by FBT0B:

$$T_{\text{rel}}^2 \hookrightarrow X_{\text{Reg}}^{(6)} \xrightarrow{\pi_{\text{fib}}} B_{\text{Reg}}^{(4)}.$$

The local splitting is defined by a connection,

$$TX_{\text{Reg}}^{(6)} = H_A \oplus V, \quad V = \text{Ker } d\pi_{\text{fib}},$$

and does not represent a global coordinate product. In the current FBT01A architecture the relative-phase vertical sector is isotropic for the six-dimensional mother symplectic form. Hence

$$\Omega_{\mathcal{B}} = \omega_H + \eta, \quad \Omega_{\mathcal{B}}|_{V \times V} = 0.$$

The mixed form η is the first-order channel through which vertical phase information can affect horizontal readout data. It constrains readability, but on the regular saturated branch it removes no vertical direction.

FBT0C v1.5 supplies a second distinction. A higher-order readout is not merely a higher-degree invariant; it may define a non-injective map from microscopic structural states to effective response data. The resulting equivalence classes are channel dependent. Thus preservation of $\Omega_{\mathcal{B}}^{\wedge k}$ under Hamiltonian flow does not imply that every lower-order microscopic representative can be reconstructed from a readout that depends only on $\Omega_{\mathcal{B}}^{\wedge k}$ or another quadratic invariant. The microscopic dynamics may remain reversible while the macroscopic readout is many-to-one.

FBT01E and FBT08A supply the Lorentzian signature and kinematical readout on the four-dimensional base. Throughout the present paper,

$$\text{sign}(g) = (-, +, +, +).$$

FBT08E constructs the effective worldline of a stable configuration. FBT15C and FBT08F separate the reduced collective zero modes from the positive transverse physical stability operator and formulate the dynamical gap required for adiabatic persistence.

FBT10A addresses the following question:

How do convergent microscopic measures, first-order mixed transmission, order-dependent readout quotients, dynamical low-energy selection, and a final macroscopic resolution combine to produce an inertial worldline coefficient and a Lorentzian response metric?

The answer is a layered architecture with three mutually distinct readout branches:

$$\begin{array}{ll}
\text{mechanical:} & \text{Morse/thimble sector} \longrightarrow \widehat{D}_i \longrightarrow M_i, \\
\text{probabilistic:} & \text{FBT06C} \longrightarrow \nu_b \longrightarrow \mu_{\text{adm}}, \\
\text{response channel } r : & \mathcal{E}_{\text{adm}} \xrightarrow{\text{P}_\eta} \mathcal{E}_{\text{read}} \xrightarrow{\Pi_{\text{inv}}^{(r)}} \mathcal{E}_{\text{inv}}^{(r)} \xrightarrow{\text{P}_{\text{ad}}^{(r)}} \mathcal{E}_{\text{low}}^{(r)} \xrightarrow{\Pi_{\text{res}}^{(r)}} \mathcal{X}_{\text{macro}}^{(r)}.
\end{array}$$

The channel-dependent composite is

$$\Pi_{\text{macro}}^{(r)} = \Pi_{\text{res}}^{(r)} \circ \text{P}_{\text{ad}}^{(r)} \circ \Pi_{\text{inv}}^{(r)} \circ \text{P}_\eta.$$

The first map is constrained by the mixed form η ; the second is the invariant-order quotient supplied by the channel readout; the third is constrained by the FBT08F gap and adiabatic regime; and the fourth is the residual macroscopic resolution. For the metric-response channel we suppress the superscript $r = \text{met}$.

The diagonal sector probabilities are conditional masses of the pushforward measure, while any surviving off-diagonal overlap data enter separately through the FBT27 coherence interface. The mechanical and metric branches meet in the low-energy worldline action

$$S_{\text{worldline}}^{(i)} = -M_i \int \sqrt{-g_{\mu\nu}^{\text{eff}} dx^\mu dx^\nu}.$$

The common appearance of structural depth does not identify the mass coefficient with a sector probability:

$$M_i \neq p_i.$$

Likewise, equality in the metric-response quotient does not imply equality in the mass readout. Distinct readout channels may impose distinct equivalence relations on the same microscopic state space.

The phrase *dual readout sectors* therefore denotes common structural origin without identity of mathematical objects. FBT10A constructs the compatible hierarchy conditionally. It does not derive a unique invariant quotient for every channel, a unique residual resolution map, a unique response-metric law, or a complete gravitational theory from the symplectic form alone.

2 Two Upstream Evolutions and the Structural Response Principle

2.1 Hamiltonian structural evolution

Let

$$H_{\text{str}} \in C^\infty(X_{\text{Reg}}^{(6)})$$

be an admissible structural Hamiltonian. Its Hamiltonian vector field is defined by

$$\iota_{X_{H_{\text{str}}}} \Omega_{\mathcal{B}} = dH_{\text{str}}.$$

The corresponding flow

$$\Phi_t : X_{\text{Reg}}^{(6)} \rightarrow X_{\text{Reg}}^{(6)}$$

preserves the mother symplectic form:

$$\Phi_t^* \Omega_{\mathcal{B}} = \Omega_{\mathcal{B}}.$$

It transports the physical carrier state and, when suitable lifted bundle automorphisms exist, the auxiliary structural data used in the readout construction.

2.2 Complex Morse and Lefschetz selection flow

The selection of a Lefschetz thimble belongs to a different dynamical category. Let

$$\mathcal{S}_{\mathbb{C}}$$

be the relevant complexified action and

$$f = \Re \mathcal{S}_{\mathbb{C}}.$$

A schematic downward Morse flow is

$$\boxed{\frac{dz}{d\lambda} = -\text{grad } f(z),}$$

with the metric and complex-conjugation conventions specified by the thimble theory. This flow organizes integration cycles and critical sectors; it is not the physical time evolution generated by $X_{H_{\text{str}}}$.

Principle FBT10A-2.1 (Two-flow separation). The FBT response hierarchy uses two logically distinct flows:

$$\boxed{\begin{array}{ll} X_{H_{\text{str}}} : & \text{physical Hamiltonian transport of structural states,} \\ -\text{grad } \Re \mathcal{S}_{\mathbb{C}} : & \text{complex-Morse selection of admissible thimbles.} \end{array}}$$

A compatible branch may be transported by the Hamiltonian flow while its semiclassical sector organization is represented by the thimble architecture. No equality of the two flows is assumed.

Assumption FBT10A-2.2 (Flow-compatibility interface). The Hamiltonian evolution preserves the admissible class of states for which the selected thimble decomposition, boundary conditions, and continuation data remain defined. Equivalently, physical evolution does not cross a wall at which the relevant complex-Morse sector ceases to be admissible unless an explicit wall-crossing analysis is supplied.

2.3 Hamiltonian Structural Response Principle

Principle FBT10A-2.3 (Hamiltonian Structural Response Principle). The primary physical evolution of the regular FBT carrier is the Hamiltonian flow Φ_t . Observable variables arise only after successive structural evaluations, tensorial descent, order-dependent readout quotienting, low-energy selection, and constitutive maps. Schematically,

$$\boxed{\begin{array}{l} (H_{\text{str}}, \Omega_{\mathcal{B}}) \longrightarrow X_{H_{\text{str}}} \longrightarrow \Phi_t \longrightarrow [\mathfrak{S}_t] \xrightarrow{\mathfrak{P}_{\text{read}}} \mathfrak{R}_{\text{adm}}(t), \\ \text{FBT06C/D/E} \longrightarrow \mu_{\text{adm}}(t) \xrightarrow{\Pi_{\text{macro}}^{(r)}} \mu_{\text{macro}}^{(r)}(t), \\ (\mathfrak{R}_{\text{adm}}(t), \{\mu_{\text{macro}}^{(r)}(t)\}_r, \rho^{\text{ov}}(t), \mathcal{C}_{\text{macro}}) \longrightarrow \mathfrak{R}_{\text{macro}}(t) \xrightarrow{\mathcal{R}_{\text{met}}} g^{\text{eff}}(t). \end{array}}$$

For each response channel,

$$\Pi_{\text{macro}}^{(r)} = \Pi_{\text{res}}^{(r)} \circ \mathsf{P}_{\text{ad}}^{(r)} \circ \Pi_{\text{inv}}^{(r)} \circ \mathsf{P}_{\eta}.$$

The thimble data enter through the independent complex-Morse sector selection and the compatible semiclassical decomposition of $[\mathfrak{S}_t]$.

Remark FBT10A-2.4 (Unity without tensor identification). The common structural ori-

gin does not assert

$$\Omega_{\text{phase}} = \eta = F_A = g^{\text{eff}} = R[g^{\text{eff}}].$$

These objects have different tensor types and live at different levels of the architecture. The unity is one of compatible structural origin and readout, not one of mathematical identity.

Principle FBT10A-2.5 (Invariant preservation versus readout reconstructibility). Hamiltonian transport preserves the symplectic wedge invariants,

$$\Phi_t^*(\Omega_B^{\wedge k}) = \Omega_B^{\wedge k},$$

but a readout depending on a higher-order invariant may be non-injective. Consequently,

$$\boxed{\text{invariant conservation} \neq \text{microscopic reconstructibility from that invariant.}}$$

The resulting loss is a loss of readout distinguishability. It does not assert microscopic dissipation or destruction of the Hamiltonian state.

2.4 Readout versus observation

The present paper constructs readout variables and constitutive maps. It does not construct experimental observation functionals. In the terminology of FBT32,

$$\boxed{\text{structural readout} \neq \text{physical observation outcome.}}$$

Mass coefficients and effective metrics are therefore called readout variables or readout sectors. A downstream map from these variables to measurement records is outside the scope of FBT10A.

3 The Response-Metric Bridge

3.1 The unique global 4 + 2 carrier architecture

Let

$$T_{\text{rel}}^2 \hookrightarrow X_{\text{Reg}}^{(6)} \xrightarrow{\pi_{\text{fib}}} B_{\text{Reg}}^{(4)}$$

be the regular principal relative-phase fibration of FBT0B. On an admissible domain

$$\mathcal{U} \subset X_{\text{Reg}}^{(6)}$$

over a Lorentzian readout domain

$$\mathcal{M}_4^{\text{st}} \subseteq B_{\text{Reg}}^{(4)},$$

the connection defines

$$T\mathcal{U} = H_A \oplus V, \quad V = \text{Ker } d\pi_{\text{fib}}, \quad \text{rank } V = 2.$$

The base bundle projection is fixed by this architecture. FBT10A does not introduce an additional point map $X_{\text{Reg}}^{(6)} \rightarrow B_{\text{Reg}}^{(4)}$.

The four-dimensional Lorentzian seed

$$g^{(0)}$$

is inherited from FBT01E and FBT08A on the selected branch. Its role is kinematic: it fixes the signature, causal cone, and the tensor category in which the constitutive metric response will be evaluated.

3.2 Isotropic vertical sector and mixed mother form

In the current FBT01A architecture the vertical relative-phase sector is isotropic for the six-dimensional mother symplectic form:

$$\Omega_{\mathcal{B}}|_{V \times V} = 0.$$

Relative to the connection splitting,

$$\Omega_{\mathcal{B}} = \omega_H + \eta,$$

where

$$\omega_H \in \Gamma(\Lambda^2 H_A^*), \quad \eta \in \Gamma(H_A^* \wedge V^*).$$

The full form remains closed and nondegenerate. In six dimensions the Liouville volume is supplied by the mixed structure rather than by a vertical symplectic block:

$$\frac{\Omega_{\mathcal{B}}^3}{3!} = \frac{\omega_H \wedge \eta^2}{2}$$

on the regular isotropic-fibre branch.

The phase curvature Ω_{phase} , the connection curvature F_A , and the readout QGT are separate structures. In particular,

$$\eta \neq A, \quad \eta \neq F_A, \quad \Omega_{\text{phase}} \neq \Omega_{\mathcal{B}}|_{V \times V}.$$

3.3 Bundle connection and lifted Hamiltonian transport

Let

$$P_{\text{rel}} \longrightarrow B_{\text{Reg}}^{(4)}$$

denote the principal T_{rel}^2 -bundle represented by $X_{\text{Reg}}^{(6)}$, with principal connection one-form

$$A \in \Omega^1(P_{\text{rel}}; \mathfrak{t}^2).$$

Since the structure group is Abelian,

$$F_A = dA.$$

The Hamiltonian flow Φ_t acts on the carrier. To transport the principal connection, one needs a lift

$$\tilde{\Phi}_t : P_{\text{rel}} \longrightarrow P_{\text{rel}}$$

covering the induced base evolution. It is therefore the lifted automorphism, not the base flow alone, that acts on the connection:

$$\tilde{\Phi}_t^* A \sim A, \quad \tilde{\Phi}_t^* F_A \sim F_A.$$

Assumption FBT10A-3.1 (Transport compatibility of the auxiliary data). The Hamiltonian flow admits the required lifts to the relevant bundles and preserves the admissible readout class. Schematically,

$$\begin{aligned} \Phi_t^*(H_A \oplus V) &\sim H_A \oplus V, \\ \tilde{\Phi}_t^* A &\sim A, \quad \tilde{\Phi}_t^* F_A \sim F_A, \end{aligned}$$

and the pulled-back Berry and readout-QGT data remain in their admissible gauge, modular, and branch-equivalence classes.

3.4 Constrained structural configuration

The structural data do not form a list of independent coordinates. For example, F_A is the curvature of A , η is determined by $\Omega_{\mathcal{B}}$ and the chosen splitting, and

$$Q^{\text{read}} = g^{\text{read}} + \frac{i}{2} F^{\text{read}}$$

contains linked real and antisymmetric components.

Definition FBT10A-3.2 (Constrained admissible structural configuration). A constrained admissible structural configuration is an equivalence class

$$[\mathfrak{S}] = \left[\Omega_{\mathcal{B}}, H_A \oplus V, \eta, A, F_A, \Omega_{\text{phase}}^{\text{eff}}, g^{\text{read}}, F^{\text{read}} \right]$$

subject to the compatibility relations

$$F_A = dA, \quad \Omega_{\mathcal{B}} = \omega_H + \eta, \quad Q^{\text{read}} = g^{\text{read}} + \frac{i}{2} F^{\text{read}},$$

and to the relevant closure, integrality, gauge, modular, and branch conditions.

The evolving structural class is denoted

$$[\mathfrak{S}_t].$$

Hamiltonian dynamics transports this constrained class; it does not create an independent evolution law for every listed entry.

3.5 Tensorial readout descent

The bundle projection

$$\pi_{\text{fib}} : X_{\text{Reg}}^{(6)} \rightarrow B_{\text{Reg}}^{(4)}$$

assigns the base point. It does not by itself convert arbitrary tensors or connections on the total space into tensors on the base. Such a conversion requires additional operations.

Definition FBT10A-3.3 (Tensorial readout-descent operator). The operator

$$\mathfrak{P}_{\text{read}}$$

denotes the admissible collection of operations that converts the constrained carrier data into response data on the Lorentzian base. It may include:

- (i) horizontal projection;
- (ii) descent of basic forms;
- (iii) basic fibrewise contraction or integration required for tensorial descent, distinct from macroscopic statistical averaging;
- (iv) transition-map and modular matching;
- (v) gauge-equivalence reduction;
- (vi) reality and Hermiticity projection for real response variables.

Thus tensorial descent produces an admissible field of real, gauge-covariant response observables,

$$[\mathfrak{S}_0] \xrightarrow{\Phi_t} [\mathfrak{S}_t] \xrightarrow{\mathfrak{P}_{\text{read}}} \mathfrak{R}_{\text{adm}}(t) \quad \text{on } \mathcal{E}_{\text{adm}}.$$

The subsequent conversion from $\mathfrak{R}_{\text{adm}}$ to $\mathfrak{R}_{\text{macro}}$ is a measure-theoretic coarse-graining operation defined below; it is not part of the bundle projection itself.

3.6 First-order mixed readability induced by η

The FBT01A mixed form does more than contribute one entry to the response package. It determines the first-order channel through which vertical relative-phase variations can affect horizontal readout data.

Definition FBT10A-3.4 (Vertical mixed-readability map). At each point of the regular bundle, define

$$\eta_V^b : V \longrightarrow H_A^*, \quad \eta_V^b(v)(h) := \eta(h, v).$$

Its kernel

$$K_\eta := \text{Ker } \eta_V^b$$

consists of vertical directions invisible to the first-order mixed symplectic pairing. The corresponding first-order readable vertical space is

$$V_{\text{read}} := V/K_\eta.$$

A vertical variation in K_η may still enter through a higher-order, nonlocal, holonomy, or different constitutive channel. The statement is narrower: it is invisible to the linear mixed pairing encoded by η .

Proposition FBT10A-3.5 (Mixed-rank saturation on the regular branch). Assume

$$\omega_H \wedge \eta^{\wedge 2} \neq 0$$

on a regular branch with $\text{rank } V = 2$. Then

$$\text{rank } \eta_V^b = 2, \quad \text{Ker } \eta_V^b = 0.$$

Hence every vertical direction is readable through the first-order mixed channel on the Liouville-saturated regular branch.

Proof. If $\text{rank } \eta_V^b \leq 1$, then locally the mixed form contains at most one independent vertical covector and can be written in decomposable form

$$\eta = \alpha \wedge \beta, \quad \alpha \in H_A^*, \quad \beta \in V^*.$$

Consequently $\eta \wedge \eta = 0$, contradicting $\omega_H \wedge \eta^{\wedge 2} \neq 0$. Since $\text{rank } V = 2$, the rank must be two and the kernel is trivial. $\square$

Remark FBT10A-3.6 (Readability is not a quadratic quotient). On the saturated regular branch, η removes no vertical direction. Its role is to determine the horizontal image

$$\text{im } \eta_V^b \subset H_A^*$$

through which vertical information is transmitted. It therefore answers which directions are geometrically readable. It does not determine whether distinct readable configurations become indistinguishable after a quadratic or higher-order response map.

Definition FBT10A-3.7 (First-order mixed-readability map). The map

$$\mathbf{P}_\eta : \mathcal{E}_{\text{adm}} \longrightarrow \mathcal{E}_{\text{read}}$$

denotes the first-order encoding of admissible microscopic data in the mixed-readable channel. At the linear vertical level it is induced by

$$V \longrightarrow V/\text{Ker } \eta_V^b$$

and by the horizontal image of η_V^b . On the saturated branch it is injective on vertical directions and should be interpreted as a transmission/readout map, not as a lossy coarse-graining.

The detailed nonlinear extension of P_η to composite observables may require the QGT, connection, and branch data contained in $[\mathfrak{S}]$. Thus η constrains the first-order readable channel without uniquely determining the higher-order response quotient or the final macroscopic observable.

3.7 Invariant-order readout quotients from FBT0C

First-order readability does not determine the resolution of a higher-order response. FBT0C v1.5 supplies the appropriate intermediate notion: a response channel may retain only an invariant or composite readout of the readable data, and that map may be non-injective.

Let $\mathfrak{C}_{\text{resp}}$ denote the family of response channels retained by the effective theory. A channel may be linear, quadratic, top-order, nonlocal, or a constitutive combination of several invariant orders.

Definition FBT10A-3.8 (Channel readout and readout equivalence). For each $r \in \mathfrak{C}_{\text{resp}}$, let

$$\mathcal{Q}^{(r)} : \mathcal{E}_{\text{read}} \longrightarrow \mathcal{J}^{(r)}$$

be the corresponding admissible response readout. Define

$$\xi_1 \sim_r \xi_2 \iff \mathcal{Q}^{(r)}(\xi_1) = \mathcal{Q}^{(r)}(\xi_2).$$

The associated invariant-order quotient is

$$\Pi_{\text{inv}}^{(r)} : \mathcal{E}_{\text{read}} \longrightarrow \mathcal{E}_{\text{inv}}^{(r)} := \mathcal{E}_{\text{read}} / \sim_r.$$

Proposition FBT10A-3.9 (Readout-quotient factorisation). *The relation $\sim_r$ is an equivalence relation. The map $\mathcal{Q}^{(r)}$ factors uniquely through the quotient:*

$$\mathcal{Q}^{(r)} = \tilde{\mathcal{Q}}^{(r)} \circ \Pi_{\text{inv}}^{(r)}.$$

Every observable of the form

$$\mathcal{O}^{(r)} = \tilde{\mathcal{O}}^{(r)} \circ \mathcal{Q}^{(r)}$$

is constant on each $\sim_r$ -equivalence class.

Proof. Equality of the readout values is reflexive, symmetric, and transitive, so it defines an equivalence relation. The universal property of the quotient gives the unique factorisation. Constancy of $\mathcal{O}^{(r)}$ on equivalence classes follows immediately. $\square$

Principle FBT10A-3.10 (Invariant-order loss of distinguishability). Whenever $\mathcal{Q}^{(r)}$ is non-injective, the corresponding response channel identifies distinct readable microscopic states. The resulting loss is

$$\boxed{\text{loss of readout distinguishability} \quad \text{or} \quad \text{non-reconstructibility},}$$

not microscopic destruction of information. The Hamiltonian state and its conserved symplectic invariants may remain well defined even when a higher-order response is many-to-one.

The FBT01A top-volume channel supplies a concrete example. With ω_H fixed, define

$$\mathcal{Q}_\eta^{\text{top}}(\eta) := \frac{1}{2} \omega_H \wedge \eta^{\wedge 2}.$$

Then

$$\mathcal{Q}_\eta^{\text{top}}(-\eta) = \mathcal{Q}_\eta^{\text{top}}(\eta).$$

Thus the sign of the first-order mixed representative is not reconstructible from this top-order density alone, although both representatives remain distinct at the linear readability level.

Remark FBT10A-3.11 (No universal loss theorem). A higher-order readout is not automatically non-injective on every admissible state space. If $\mathcal{Q}^{(r)}$ is injective, then the quotient is lossless and may be identified with the identity map. The additional quotient layer is therefore channel dependent and appears only when the actual response law is non-injective.

Principle FBT10A-3.12 (Channel dependence of readout equivalence). Different response channels may define different equivalence relations on the same microscopic state space. In particular,

$$\xi_1 \sim_{\text{met}} \xi_2$$

need not imply equality of the mechanical mass readout,

$$M(\xi_1) = M(\xi_2).$$

Mass and metric response are therefore dual readout channels precisely because they can retain different structural distinctions.

3.8 Adiabatic low-energy selection from FBT08F

Invariant-order distinguishability does not imply dynamical persistence. A readable invariant class may contain modes whose representatives oscillate on a timescale too short to survive in the collective low-energy description.

FBT15C and FBT08F supply the reduced collective bundle and the positive transverse dynamical gap. At the linearised level, let

$$P_{\text{coll},q}$$

denote the smooth spectral projection onto the collective zero-mode bundle. Let

$$\Omega_q := \left(\mathcal{K}_q^{-1/2} \mathcal{L}_{\text{stab},q} \mathcal{K}_q^{-1/2} \right)^{1/2}$$

be the positive transverse frequency operator. Choose a low-energy threshold Λ_{IR} satisfying

$$0 \leq \Lambda_{\text{IR}} < \omega_{\text{gap}}.$$

With the FBT08F gap convention, no transverse eigenfrequency lies below this threshold, so the raw low-energy projector is $P_{\text{coll},q}$. If a downstream model designates additional soft modes as retained variables, the retained bundle must first be enlarged and ω_{gap} redefined as the gap above that enlarged bundle.

The order-dependent quotient and the dynamical selector must be compatible.

Assumption FBT10A-3.13 (Quotient-adiabatic compatibility). For each retained response channel r , the raw low-energy projection preserves the channel equivalence relation:

$$\xi_1 \sim_r \xi_2 \implies P_{\text{low}} \xi_1 \sim_r P_{\text{low}} \xi_2.$$

Consequently it descends to a well-defined measurable map

$$\boxed{\mathbf{P}_{\text{ad}}^{(r)} : \mathcal{E}_{\text{inv}}^{(r)} \longrightarrow \mathcal{E}_{\text{low}}^{(r)}, \quad \mathbf{P}_{\text{ad}}^{(r)}[\xi]_r := [P_{\text{low}} \xi]_r.}$$

All modes excluded by $\mathbf{P}_{\text{ad}}^{(r)}$ are separated from the retained sector by a uniform positive frequency gap.

For the selected metric-response channel we write simply

$$P_{\text{ad}} : \mathcal{E}_{\text{inv}} \rightarrow \mathcal{E}_{\text{low}}.$$

Proposition FBT10A-3.14 (Adiabatic suppression after the readout quotient). *Assume the FBT08F propagator and regularity hypotheses, the quotient–adiabatic compatibility above, and let $\mathcal{O}^{(r)}$ be a smooth locally Lipschitz observable that factors through $\mathcal{E}_{\text{inv}}^{(r)}$. Then, through every interval on which the gap remains open,*

$$\mathcal{O}^{(r)}([\Psi]_r) - \mathcal{O}^{(r)}(P_{\text{ad}}^{(r)}[\Psi]_r) = O(\varepsilon_{\text{ad}})$$

for an adiabatically prepared branch, up to the model-dependent order of the derivative expansion.

Proof. FBT08F gives a transverse displacement of order $O(\varepsilon_{\text{ad}})$ from the collective or retained low-energy family. Compatibility makes the projected class independent of the representative, and local Lipschitz continuity converts the displacement estimate into the displayed observable estimate. $\square$

Thus P_η determines first-order transmission, $\Pi_{\text{inv}}^{(r)}$ determines the distinctions retained by response channel r , and $P_{\text{ad}}^{(r)}$ determines which of those distinctions survive at the collective timescale.

3.9 Channel-dependent factorised macroscopic coarse-graining

A final resolution choice is still required after first-order readability, invariant-order quotienting, and dynamical low-energy selection have been imposed.

Definition FBT10A-3.15 (Residual macroscopic resolution map). For each response channel r , a residual macroscopic resolution map is a measurable map

$$\Pi_{\text{res}}^{(r)} : \mathcal{E}_{\text{low}}^{(r)} \longrightarrow \mathcal{X}_{\text{macro}}^{(r)}$$

which identifies low-energy invariant classes that remain unresolved by the selected macroscopic description.

Definition FBT10A-3.16 (Channel-dependent macroscopic coarse-graining map). For $r \in \mathfrak{C}_{\text{resp}}$, define

$$\Pi_{\text{macro}}^{(r)} := \Pi_{\text{res}}^{(r)} \circ P_{\text{ad}}^{(r)} \circ \Pi_{\text{inv}}^{(r)} \circ P_\eta : \mathcal{E}_{\text{adm}} \longrightarrow \mathcal{X}_{\text{macro}}^{(r)}.$$

Equivalently,

$$\mathcal{E}_{\text{adm}} \xrightarrow{P_\eta} \mathcal{E}_{\text{read}} \xrightarrow{\Pi_{\text{inv}}^{(r)}} \mathcal{E}_{\text{inv}}^{(r)} \xrightarrow{P_{\text{ad}}^{(r)}} \mathcal{E}_{\text{low}}^{(r)} \xrightarrow{\Pi_{\text{res}}^{(r)}} \mathcal{X}_{\text{macro}}^{(r)}.$$

For the selected metric-response channel $r = \text{met}$, the superscript is suppressed:

$$\Pi_{\text{inv}} := \Pi_{\text{inv}}^{(\text{met})}, \quad P_{\text{ad}} := P_{\text{ad}}^{(\text{met})}, \quad \Pi_{\text{res}} := \Pi_{\text{res}}^{(\text{met})}, \quad \Pi_{\text{macro}} := \Pi_{\text{macro}}^{(\text{met})}.$$

Principle FBT10A-3.17 (Readability–quotient–persistence–resolution separation). The four factors of $\Pi_{\text{macro}}^{(r)}$ have distinct logical roles:

P_η	first-order geometric readability determined by η ,
$\Pi_{\text{inv}}^{(r)}$	channel-dependent invariant/readout equivalence,
$P_{\text{ad}}^{(r)}$	dynamical persistence determined by the low-energy gap,
$\Pi_{\text{res}}^{(r)}$	final macroscopic resolution and constitutive identification.

Only information that is readable, distinguished by the chosen response channel, dynamically persistent, and finally resolved reaches the macroscopic response.

The factorisation reduces but does not eliminate constitutive freedom. The form η constrains first-order transmission; FBT0C constrains the quotient once the response map $\mathcal{Q}^{(r)}$ is specified; the spectral gap constrains the dynamical selector; and $\Pi_{\text{res}}^{(r)}$ remains the residual resolution input. In particular,

$$\eta \not\Rightarrow \Pi_{\text{macro}}^{(r)} \quad \text{uniquely.}$$

Surviving FBT27 overlap data must pass through the same channel quotient, low-energy selector, and residual resolution. Coherence that is erased by any of these maps does not enter the corresponding macroscopic response.

3.10 Macroscopic structural averaging from the assembled admissible measure

The probabilistic structure used by FBT10A is inherited from the FBT06 convergence sequence rather than postulated at the response-metric level.

For every regular base point $b \in B_{\text{Reg}}^{(4)}$, FBT06D constructs a normalized probability measure on the unique admissible fibrewise thimble,

$$d\nu_b(\phi) = \frac{1}{Z_b} \exp[-\mathcal{D}_b(\phi)] d\mathbf{m}_{\text{fract},b}(\phi), \quad \nu_b(\mathcal{J}_*(b)) = 1.$$

FBT06E assembles these probability kernels over the regular base. We denote the resulting measure on the admissible measurable total space by μ_{adm} , with the disintegration formula

$$\int_{\mathcal{E}_{\text{adm}}} F d\mu_{\text{adm}} = \int_{B_{\text{Reg}}^{(4)}} \left(\int_{\mathcal{J}_*(b)} F(b, \phi) d\nu_b(\phi) \right) d\mu_{B_{\text{Reg}}^{(4)}}(b).$$

Here $\mu_{B_{\text{Reg}}^{(4)}}$ is the selected σ -finite reference measure on the regular base. The present paper uses only this measure assembly; it does not depend on any separate claim concerning direct-integral Hilbert–Schmidt operators.

Principle FBT10A-3.18 (No second probability postulate). FBT10A does not convert a complex thimble amplitude into a probability by an additional map. Absolute convergence is supplied by FBT06C, fibrewise probability normalization by FBT06D, and measurable assembly by FBT06E. FBT10A begins only at the later stage of macroscopic resolution and conditional averaging.

The factorised map of Definition FBT10A-3.16 induces the macroscopic probability measure

$$\mu_{\text{macro}} := (\Pi_{\text{macro}})_* \mu_{\text{adm}} = (\Pi_{\text{res}})_* (\mathbf{P}_{\text{ad}})_* (\Pi_{\text{inv}})_* (\mathbf{P}_{\eta})_* \mu_{\text{adm}}.$$

Thus the pushforward is constrained successively by first-order mixed readability, metric-channel invariant quotienting, adiabatic low-energy selection, and residual macroscopic resolution. Let $\mathcal{O}$ be an integrable real gauge-covariant response observable on $\mathcal{E}_{\text{adm}}$. Its macroscopic readout is represented by the conditional expectation

$$\langle \mathcal{O} \rangle_{\text{macro}} := \mathbb{E}_{\mu_{\text{adm}}} [\mathcal{O} \mid \sigma(\Pi_{\text{macro}})]. \quad (1)$$

Whenever μ_{adm} admits a disintegration over Π_{macro} ,

$$\mu_{\text{adm}} = \int_{\mathcal{X}_{\text{macro}}} \mu_x d\mu_{\text{macro}}(x),$$

this becomes

$$\langle \mathcal{O} \rangle_{\text{macro}}(x) = \int_{\Pi_{\text{macro}}^{-1}(x)} \mathcal{O}(\xi) d\mu_x(\xi). \quad (2)$$

3.10.1 Discrete-sector reduction

Suppose that at the chosen resolution the conditional fibre $\Pi_{\text{macro}}^{-1}(x)$ decomposes into a finite or countable family of resolved admissible sectors $\mathcal{S}_i(x)$. Define

$$\boxed{p_i(x) := \mu_x(\mathcal{S}_i(x)).} \quad (3)$$

Then

$$p_i(x) \geq 0, \quad \sum_i p_i(x) = 1.$$

If the observable is effectively constant on each sector, $\mathcal{O}|_{\mathcal{S}_i(x)} \simeq \mathcal{O}_i(x)$, then

$$\boxed{\langle \mathcal{O} \rangle_{\text{macro}}(x) = \sum_i p_i(x) \mathcal{O}_i(x).} \quad (4)$$

The quantities p_i are induced conditional probabilities, not independent constitutive weights.

3.10.2 Within-sector damping and sector-level depth

Two uses of the word *depth* must be distinguished. The function

$$\mathcal{D}_i(\phi), \quad \phi \in \mathcal{J}_{*,i},$$

is the within-sector damping potential entering the FBT06D measure. The number $\widehat{D}_i$ is a chosen lifted critical or branch depth used in the comparison of distinct stable sectors. Thus

$$\boxed{\mathcal{D}_i(\phi) \neq \widehat{D}_i.}$$

A saddle approximation may relate the local density around sector i to $e^{-\widehat{D}_i}$, but the complete sector probability is the integrated measure

$$p_i(x) = \mu_x(\mathcal{S}_i(x)),$$

not the isolated exponential factor.

More explicitly, on a measurable sector one may write a Radon–Nikodym representation

$$d\mu_x|_{\mathcal{S}_i} = r_i(\xi) d\lambda_i(\xi), \quad r_i \geq 0.$$

In a leading semiclassical chart,

$$r_i^{\text{sc}}(\xi) = Z_i^{\text{meas}}(\xi) \exp[-\mathcal{D}_i(\xi)], \quad Z_i^{\text{meas}}(\xi) \geq 0,$$

may approximate this density. A complex saddle coefficient such as

$$\mathfrak{a}_i = n_i \Delta_i e^{-\widehat{D}_i} \mathcal{C}_i$$

may organize a semiclassical expansion, but it is not itself identified with the positive probability p_i .

Remark FBT10A-3.19 (Averaging preserves selected global data). Coarse-graining need not erase affine levels, Berry periods, monodromy, holonomy, thimble classes, or branch labels. It removes only those microscopic distinctions that are unresolved by the selected macroscopic readout.

3.11 Diagonal probabilities and FBT27 overlap corrections

The probability measure inherited from FBT06D–FBT06E determines the diagonal population of resolved sectors. It does not by itself imply that every pair of distinct sectors has become fully decoherent.

In the sector-diagonal regime,

$$\langle \mathcal{O} \rangle_{\text{macro}}^{\text{diag}} = \sum_i p_i \mathcal{O}_{ii}.$$

If an FBT27-compatible overlap construction survives the selected invariant quotient, adiabatic filter, and residual resolution, the response may contain additional off-diagonal terms,

$$\langle \mathcal{O} \rangle_{\text{macro}} = \sum_i p_i \mathcal{O}_{ii} + \sum_{i \neq j} \rho_{ij}^{\text{ov}} \mathcal{O}_{ji}. \quad (5)$$

The coefficients ρ_{ij}^{ov} encode effective coherence or overlap after readout descent. They are not sector probabilities. When a reduced density description exists, the combined matrix must satisfy

$$\rho^{\text{eff}} = (\rho_{ij}^{\text{eff}}) \geq 0, \quad \text{Tr } \rho^{\text{eff}} = 1, \quad \rho_{ii}^{\text{eff}} = p_i,$$

with $\rho_{ij}^{\text{ov}} = \rho_{ij}^{\text{eff}}$ for $i \neq j$.

Principle FBT10A-3.20 (Probability–overlap separation). Diagonal sector probabilities and inter-sector overlaps are distinct readout data:

$$\begin{array}{ll} p_i : & \text{conditional probabilities inherited from FBT06D–FBT06E,} \\ \rho_{ij}^{\text{ov}} : & \text{off-diagonal coherence constrained by the FBT27 sequence.} \end{array}$$

An overlap coefficient must not be absorbed into a classical probability unless a specific decoherence theorem justifies that reduction.

In the fully decohered limit,

$$\rho_{ij}^{\text{ov}} \longrightarrow 0 \quad (i \neq j),$$

and equation (5) reduces to the ordinary probability average. Conversely, surviving overlap data enter the macroscopic response package without changing $\sum_i p_i = 1$.

Remark FBT10A-3.21 (Division of analytic responsibilities). The relevant division is

$$\begin{array}{ll} \text{FBT06D} : & \text{within-thimble probability normalization,} \\ \text{FBT06E} : & \text{measurable assembly of the probability structure,} \\ \text{FBT27} : & \text{inter-sector overlap, transition, and coherence data,} \\ \text{FBT10A} : & \text{macroscopic coarse-graining and constitutive response.} \end{array}$$

3.12 Admissible macroscopic response-data space

A representative real response package may be written schematically as

$$\mathfrak{R}_{\text{macro}} = (\langle \Omega_{\text{phase}}^{\text{eff}} \rangle_{\text{macro}}, \langle \eta \rangle_{\text{macro}}, [A]_{\text{gauge}}, \langle F_A \rangle_{\text{macro}}, \langle g^{\text{read}} \rangle_{\text{macro}}, \langle F^{\text{read}} \rangle_{\text{macro}}; \mu_{\text{macro}}, \rho^{\text{ov}}, \mathcal{C}_{\text{macro}}).$$

Here

$$\mu_{\text{macro}} = (\Pi_{\text{res}})_*(\mathbf{P}_{\text{ad}})_*(\Pi_{\text{inv}})_*(\mathbf{P}_{\eta})_*\mu_{\text{adm}}$$

records the coarse-grained probability structure inherited from FBT06D–FBT06E after first-order readability, invariant-order quotienting, adiabatic selection, and residual resolution. The tensor ρ^{ov} denotes any inter-sector coherence data that survive the same filters and are inherited from an FBT27-compatible overlap construction. In a fully decohered sector-diagonal regime,

$$\rho^{\text{ov}} = 0.$$

The complex QGT is not inserted into a real metric map as an unrestricted complex tensor. Its components enter through real Hermitian or gauge-covariant combinations. Likewise, μ_{macro} and ρ^{ov} are constrained by positivity, normalization, and compatibility with the selected coarse-graining.

Definition FBT10A-3.22 (Constrained admissible response-data space). The space

$$\mathfrak{X}_{\text{macro}}^{\text{adm}}$$

consists of real response packages satisfying:

- (i) the structural compatibility constraints inherited from $[\mathfrak{S}]$;
- (ii) gauge and modular covariance;
- (iii) reality and Hermiticity;
- (iv) positivity and normalization of μ_{macro} ;
- (v) positivity and trace compatibility of any retained coherence matrix ρ^{ov} ;
- (vi) compatibility with the first-order readability map $\mathbf{P}_{\eta}$, the metric-channel quotient Π_{inv} , the adiabatic selector $\mathbf{P}_{\text{ad}}$, and the residual resolution map Π_{res} ;
- (vii) regularity and low-energy locality conditions;
- (viii) admissible branch and boundary conditions;
- (ix) the integrability conditions required for any claimed variational completion.

Thus $\mathfrak{R}_{\text{macro}} \in \mathfrak{X}_{\text{macro}}^{\text{adm}}$, but its entries are not independent unconstrained fields.

3.13 Stable configurations and the corrected FBT08F interface

Let

$$\mathfrak{C}_q, \quad q \in \mathcal{M}_{\text{coll}},$$

be a smooth family of admissible stable internal configurations. The collective coordinates q^a belong to the internal moduli problem; their downstream Lorentzian image is the effective worldline

$$X : I \rightarrow \mathcal{M}_4^{\text{st}}.$$

The reduced Hessian of FBT15C and FBT08F retains the physical collective zero modes:

$$\boxed{\text{Ker } \mathcal{H}_{\text{red},q} = \text{span} \{ \partial_{q^a} \mathfrak{C}_q \}_{a=1}^{d_{\text{coll}}}.$$

The positive physical stability operator is the restriction to the normal bundle,

$$\mathcal{L}_{\text{stab},q} := \mathcal{H}_{\text{red},q}|_{\mathcal{N}_q},$$

and satisfies

$$\mathcal{L}_{\text{stab},q} \geq \Delta_{\text{int}} > 0, \quad \text{Ker } \mathcal{L}_{\text{stab},q} = \{0\}.$$

The four worldline coordinates X^μ are not asserted to be four normalizable zero modes of a fixed-time internal Hessian.

Let $\mathcal{K}_q$ be the positive kinetic operator controlling the linearized transverse dynamics. The dynamical frequency gap is

$$\omega_{\text{gap}}^2 := \inf \text{Spec}_+ \left(\mathcal{K}_q^{-1/2} \mathcal{L}_{\text{stab},q} \mathcal{K}_q^{-1/2} \right).$$

The adiabatic parameter is

$$\varepsilon_{\text{ad}} := \frac{\Omega_{\text{ext}}}{\omega_{\text{gap}}} \ll 1.$$

The static gap Δ_{int} may bound ω_{gap} only after the kinetic normalization has been specified.

The smooth collective/low-energy spectral projection determined by this architecture is the detailed dynamical input behind the adiabatic selector $\mathbf{P}_{\text{ad}}$ introduced in Section 3.8.

Principle FBT10A-3.23 (Stable worldline-EFT interface). Assume the collective-coordinate construction of FBT08E. Assume also the reduced Morse–Bott and positive-gap architecture of FBT15C and FBT08F, a smooth bounded transverse propagator, timelike branch selection, reparametrization invariance, and a local worldline effective theory. Then the leading zero-derivative term of the adiabatic action has the form

$$S_{\text{part}}[X, g^{\text{eff}}; \mathfrak{C}] = -M_{\mathfrak{C}} \int ds_{\text{eff}} + S_{\text{int,corr}},$$

where

$$S_{\text{int,corr}} = \sum_{n \geq 1} \varepsilon_{\text{ad}}^n S_{(n)}$$

up to model-dependent derivative counting.

3.14 Off-shell metric variation and source definition

The constitutive relation

$$g^{\text{eff}} = g^{(0)} + \mathcal{R}_{\text{met}}(\mathfrak{R}_{\text{macro}})$$

should not be imposed inside the metric variation before a common effective action has been specified. For the variational definition of the source, g^{eff} is promoted off shell to an independent low-energy field.

Convention FBT10A-3.24 (Off-shell metric and fixed-branch mass). In the variation defining the point-particle stress tensor, the branch label and the matched coefficient $M_{\mathfrak{C}}$ are held fixed:

$$\left. \frac{\delta M_{\mathfrak{C}}}{\delta (g^{\text{eff}})^{\mu\nu}} \right|_{\text{fixed branch}} = 0.$$

If a downstream model makes $M_{\mathfrak{C}}$ an explicit functional of the metric or response data, the corresponding chain-rule contributions must be added to the source tensor.

The variational source is

$$T_{\mu\nu}^{\mathfrak{C}}(x) := - \frac{2}{\sqrt{-g^{\text{eff}}(x)}} \frac{\delta S_{\text{part}}}{\delta (g^{\text{eff}})^{\mu\nu}(x)}.$$

For the leading point-particle term and the fixed-branch convention,

$$T_{\mathfrak{C}}^{\mu\nu}(x) = M_{\mathfrak{C}} \int ds_{\text{eff}} u^\mu u^\nu \frac{\delta^{(4)}(x - X(s_{\text{eff}}))}{\sqrt{-g^{\text{eff}}(x)}} + T_{\text{int}}^{\mu\nu}(x),$$

with

$$g_{\mu\nu}^{\text{eff}} u^\mu u^\nu = -1$$

in the $(-+++)$ convention. The correction tensor may contain finite-size, polarization, QGT, and spin contributions.

3.15 Particle response and the no-double-counting condition

The same stable configuration may perturb the local structural response data:

$$\delta\mathfrak{D}_{\text{resp}}^{\mathfrak{C}} = \left(\delta\eta, \delta F_A, \delta g^{\text{read}}, \delta F^{\text{read}}, \delta\Omega_{\text{phase}}^{\text{eff}}, \delta\mu_{\text{adm}}, \delta\rho^{\text{ov}} \right),$$

which, at fixed readability, adiabatic, and resolution maps, yields

$$\delta\mu_{\text{macro}}^{\mathfrak{C}} = (\Pi_{\text{macro}})_* \delta\mu_{\text{adm}}^{\mathfrak{C}}.$$

If the configuration also changes η , the selected invariant readout, the dynamical projector, or the chosen macroscopic resolution, the full linearisation contains the additional variation of

$$\Pi_{\text{macro}} = \Pi_{\text{res}} \circ \mathbf{P}_{\text{ad}} \circ \Pi_{\text{inv}} \circ \mathbf{P}_\eta.$$

The response package then includes both the measure variation and the filter-map variation. In either case,

$$\delta\mathfrak{R}_{\mathfrak{C}} = \text{CG}_{\Pi_{\text{macro}}} \left(\mathfrak{P}_{\text{read}}[\delta\mathfrak{D}_{\text{resp}}^{\mathfrak{C}}]; \delta\mu_{\text{macro}}^{\mathfrak{C}}, \delta\rho_{\mathfrak{C}}^{\text{ov}}, \delta\mathcal{C}_{\text{macro}}^{\mathfrak{C}} \right),$$

Around a background response package $\mathfrak{R}_{\text{macro}}^{(0)}$,

$$\delta g_{\mu\nu}^{\mathfrak{C}} = D\mathcal{R}_{\text{met}}|_{\mathfrak{R}_{\text{macro}}^{(0)}} [\delta\mathfrak{R}_{\mathfrak{C}}]_{\mu\nu} + O(\|\delta\mathfrak{R}_{\mathfrak{C}}\|^2).$$

Principle FBT10A-3.25 (Source–response matching principle). The arrows

$$\mathfrak{C} \longrightarrow T_{\mu\nu}^{\mathfrak{C}}$$

and

$$\mathfrak{C} \longrightarrow \delta\mathfrak{R}_{\mathfrak{C}} \longrightarrow \delta g_{\mu\nu}^{\mathfrak{C}}$$

are two candidate macroscopic descriptions of related information. They are subject to a downstream consistency condition and must not be inserted as independent duplicate sources. A complete theory must either:

- (i) derive both from one diffeomorphism-covariant effective action; or
- (ii) take one as primary and match the other through an explicit constitutive equation.

FBT10A establishes the interface but does not prove the complete matching.

3.16 The constitutive response-metric map

The macroscopic response package is not yet a metric. An additional branch-dependent constitutive assignment is required.

Definition FBT10A-3.26 (Response-metric map). A response-metric map is a real, gauge-independent, coordinate-covariant assignment

$$\mathcal{R}_{\text{met}} : \mathfrak{X}_{\text{macro}}^{\text{adm}} \longrightarrow \Gamma\left(\text{Sym}^2 T^* B_{\text{Reg}}^{(4)}\right).$$

For $\mathfrak{R}_{\text{macro}} \in \mathfrak{X}_{\text{macro}}^{\text{adm}}$, define

$$\delta g := \mathcal{R}_{\text{met}}(\mathfrak{R}_{\text{macro}}), \quad g^{\text{eff}} = g^{(0)} + \delta g.$$

The notation δg does not by itself impose perturbative smallness. Smallness is required only when signature control or the constitutive expansion is established perturbatively.

Criterion FBT10A-3.27 (Admissible response-metric branch). A response-metric branch is admissible on an open set $U \subset B_{\text{Reg}}^{(4)}$ if:

- (i) δg is a smooth real symmetric tensor;
- (ii) $g^{\text{eff}} = g^{(0)} + \delta g$ is nondegenerate;
- (iii) g^{eff} has signature $(-+++)$;
- (iv) its causal cone is compatible with the FBT08A propagation readout;
- (v) the construction is covariant under base-coordinate transitions;
- (vi) the output is independent of local gauge representatives;
- (vii) the output is compatible with admissible modular changes of the relative-phase cycle basis;
- (viii) the response data satisfy the constrained compatibility relations of $\mathfrak{X}_{\text{macro}}^{\text{adm}}$;
- (ix) the matched branch admits a diffeomorphism-covariant effective variational completion whenever a downstream action principle is claimed.

Remark FBT10A-3.28 (Why the map is constitutive). The mother symplectic form is antisymmetric and lives on a six-dimensional total space, whereas the desired response is a symmetric tensor on the four-dimensional base. A compatible almost-complex structure can produce a six-dimensional metric $g_J(u, v) = \Omega_{\mathcal{B}}(u, Jv)$, but it does not canonically determine the selected four-dimensional Lorentzian response. Readout descent, symmetric contraction data, constitutive coefficients, and branch matching remain necessary.

Proposition FBT10A-3.29 (No canonical Lorentzian response from the mother form). *The data $(X_{\text{Reg}}^{(6)}, \Omega_{\mathcal{B}})$ alone do not canonically determine a Lorentzian response metric on $B_{\text{Reg}}^{(4)}$.*

Proof. The desired construction changes both the tensor type and the geometric space. Neither the descent from the total space to the base nor the conversion of an antisymmetric two-form into a real symmetric tensor is fixed by $\Omega_{\mathcal{B}}$ alone. The additional data listed in Criterion FBT10A-3.27 are therefore necessary. $\square$

3.17 Readout covariance and global descent

Let local response descriptions on base charts U_α and U_β be related by a coordinate transition

$$h_{\alpha\beta} : U_{\alpha\beta} \rightarrow U_{\beta\alpha}.$$

Assume the constrained response packages and constitutive maps obey

$$\mathcal{R}_{\text{met},\alpha}(\mathfrak{R}_{\text{macro},\alpha}) = h_{\alpha\beta}^* \mathcal{R}_{\text{met},\beta}(\mathfrak{R}_{\text{macro},\beta}),$$

and

$$g_\alpha^{(0)} = h_{\alpha\beta}^* g_\beta^{(0)}.$$

Then

$$g_\alpha^{\text{eff}} = h_{\alpha\beta}^* g_\beta^{\text{eff}}.$$

Proposition FBT10A-3.30 (Global response-metric descent). *If the readout descent and the response-metric map are equivariant under all admissible chart, gauge, and modular transitions, the locally constructed tensors patch to a globally defined Lorentzian metric on the selected base branch.*

Proof. The overlap equation is the standard cocycle compatibility condition for a section of $\text{Sym}^2 T^* B_{\text{Reg}}^{(4)}$. The admissibility criterion preserves the Lorentzian signature. $\square$

3.18 Time dependence and variational promotion

The transported structural class gives the descended admissible response field

$$\mathfrak{R}_{\text{adm}}(t) = \mathfrak{P}_{\text{read}}([\mathfrak{S}_t]).$$

The FBT06D–FBT06E probability architecture supplies the corresponding assembled measure $\mu_{\text{adm}}(t)$. The mixed form, the selected response functional, the dynamical spectrum, and the residual resolution may all vary along the admissible branch. For each retained channel,

$$\Pi_{\text{macro}}^{(r)}(t) = \Pi_{\text{res}}^{(r)}(t) \circ \mathsf{P}_{\text{ad}}^{(r)}(t) \circ \Pi_{\text{inv}}^{(r)}(t) \circ \mathsf{P}_\eta(t).$$

For the metric channel,

$$\mu_{\text{macro}}(t) = (\Pi_{\text{macro}}(t))_* \mu_{\text{adm}}(t).$$

The response package is built from

$$\mathfrak{R}_{\text{macro}}(t) = \left(\mathbb{E}_{\mu_{\text{adm}}(t)}[\mathfrak{R}_{\text{adm}}(t) \mid \sigma(\Pi_{\text{macro}}(t))]; \mu_{\text{macro}}(t), \rho^{\text{ov}}(t), \mathcal{C}_{\text{macro}} \right).$$

Only overlap data that remain distinguished by $\Pi_{\text{inv}}(t)$, lie in the image of $\mathsf{P}_{\text{ad}}(t)$, and are resolved by $\Pi_{\text{res}}(t)$ contribute to $\rho^{\text{ov}}(t)$.

The constitutive evaluation is

$$g^{\text{eff}}(t) = g^{(0)} + \mathcal{R}_{\text{met}}(\mathfrak{R}_{\text{macro}}(t)).$$

This is not yet an independent field equation. A downstream metric action requires an off-shell promotion of g^{eff} , together with a matching equation that relates its on-shell solution to the constitutive response.

Remark FBT10A-3.31 (Two meanings of dynamics). The microscopic carrier flow is generated by H_{str} . The invariant quotient is fixed by the selected response channel, the adiabatic selector is determined by the instantaneous reduced dynamical spectrum, and the low-energy metric evolution, when present, is generated by a separate diffeomorphism-covariant effective action. The response architecture connects these layers but does not identify their equations of motion.

3.19 A quadratic real response channel and its readout quotient

Let

$$\mathcal{K} \in \Omega^2(B_{\text{Reg}}^{(4)}; \mathfrak{t}^2)$$

be a descended relative-phase curvature satisfying

$$\pi_{\text{fib}}^* \mathcal{K} = F_A$$

on the regular basic sector. Choose a symmetric positive pairing $\langle \cdot, \cdot \rangle_V$ on $\mathfrak{t}^2$ and a symmetric horizontal contraction tensor $H^{\alpha\beta}$. Define

$$Q_{\mu\nu}[\mathcal{K}] := \langle \mathcal{K}_{\mu\alpha}, \mathcal{K}_{\nu\beta} \rangle_V H^{\alpha\beta}.$$

Proposition FBT10A-3.32 (Symmetry and sign blindness of the quadratic channel). *The tensor $Q_{\mu\nu}[\mathcal{K}]$ is real and symmetric. Moreover,*

$$Q_{\mu\nu}[-\mathcal{K}] = Q_{\mu\nu}[\mathcal{K}].$$

Proof. The Lie-algebra indices are paired and the contracted base indices are summed with a symmetric tensor. Exchanging μ and ν , then interchanging the dummy indices α and β , leaves the expression unchanged. Replacing $\mathcal{K}$ by $-\mathcal{K}$ introduces two minus signs, whose product is positive. $\square$

Definition FBT10A-3.33 (Quadratic metric-response equivalence). For admissible curvature configurations, define

$$\mathcal{K}_1 \sim_Q \mathcal{K}_2 \iff Q[\mathcal{K}_1] = Q[\mathcal{K}_2].$$

The corresponding quotient map is denoted

$$\Pi_Q : \mathcal{H}_{\text{adm}} \longrightarrow \mathcal{H}_{\text{adm}} / \sim_Q.$$

Proposition FBT10A-3.34 (Quadratic response factorisation). *Every constitutive metric contribution that depends on $\mathcal{K}$ only through $Q[\mathcal{K}]$ factors through Π_Q . Consequently, microscopic curvature configurations in the same $\sim_Q$ -class are not distinguishable by that contribution to the metric response.*

Proof. This is Proposition FBT10A-3.9 applied to the channel map $\mathcal{Q}^{(\text{met})} = Q$. $\square$

The relation $\mathcal{K} \sim_Q -\mathcal{K}$ is the simplest guaranteed identification. Further degeneracies may occur depending on the vertical pairing, the contraction tensor, symmetries, and the admissible configuration space. This is a concrete FBT10A realisation of the FBT0C principle that a non-injective quadratic readout induces a natural loss of readout distinguishability.

A local constitutive ansatz may contain

$$\delta g_{\mu\nu} = \lambda_1 Q_{\mu\nu} + \lambda_2 \rho_{\text{mix}} Q_{\mu\nu} + \lambda_3 S_{\mu\nu}^{\text{QGT}} + \dots,$$

where the QGT contribution is built from real Hermitian combinations and the ellipsis denotes holonomy, derivative, nonlocal, and higher-mode terms. The metric-channel quotient $\Pi_{\text{inv}}^{(\text{met})}$ is the common refinement of the equivalence relations generated by the actual response terms retained in the model. If a term remains linear and injective on its admissible domain, its individual invariant quotient is the identity. The displayed ansatz is not a uniqueness theorem.

3.20 Central S -sector specialization

FBT12C supplies the central zero-mode response tuple

$$\mathfrak{R}_S = \left(F^S, k_S^{\text{aff}}, \Omega_{\text{phase}}, \omega_{\text{mix}} \right).$$

The local curvature Ω_{phase} and its integral level

$$k_S^{\text{aff}} = \frac{1}{2\pi} \int_{\Sigma_2} \Omega_{\text{phase}}$$

are complementary data: the former is local curvature information, while the latter records the selected global affine branch.

Let

$$\mathfrak{X}_S^{\text{adm}}$$

be the constrained central response-data space.

Definition FBT10A-3.35 (Central specialization map). The map

$$\iota_S : \mathfrak{X}_S^{\text{adm}} \longrightarrow \mathfrak{X}_{\text{macro}}^{\text{adm}}$$

identifies the central response tuple with its compatible sector in the general macroscopic response-data space.

The central constitutive metric map is

$$\mathcal{C}_{\text{met}} = \mathcal{R}_{\text{met}} \circ \iota_S.$$

Thus

$$\delta g^{(S)} = \mathcal{C}_{\text{met}}(F^S, k_S^{\text{aff}}, \Omega_{\text{phase}}, \omega_{\text{mix}}).$$

FBT12D applies this specialization within a conditional Palatini realization. It does not introduce a second independent metric-emergence mechanism.

3.21 Structural construction of the response metric

Construction FBT10A-3.36 (Invariant-adiabatic response-metric construction). Assume:

- (i) the FBT0B regular relative-phase fibration;
- (ii) the FBT01A isotropic vertical sector and mixed mother form;
- (iii) Hamiltonian structural evolution and its admissible bundle lifts;
- (iv) an independently defined complex-Morse/thimble sector compatible with that evolution;
- (v) a real tensorial readout descent $\mathfrak{P}_{\text{read}}$;
- (vi) the normalised fibre measures of FBT06D and their measurable assembly μ_{adm} from FBT06E;
- (vii) the first-order mixed-readability map

$$\mathbf{P}_\eta : \mathcal{E}_{\text{adm}} \rightarrow \mathcal{E}_{\text{read}}$$

induced by η ;

- (viii) for every retained response channel r , an admissible channel map $\mathcal{Q}^{(r)}$ and its FBT0C readout quotient

$$\Pi_{\text{inv}}^{(r)} : \mathcal{E}_{\text{read}} \rightarrow \mathcal{E}_{\text{inv}}^{(r)};$$

- (ix) quotient–adiabatic compatibility and a smooth induced selector

$$\mathbf{P}_{\text{ad}}^{(r)} : \mathcal{E}_{\text{inv}}^{(r)} \rightarrow \mathcal{E}_{\text{low}}^{(r)}$$

compatible with the FBT08F dynamical gap;

- (x) measurable residual resolution maps

$$\Pi_{\text{res}}^{(r)} : \mathcal{E}_{\text{low}}^{(r)} \rightarrow \mathcal{X}_{\text{macro}}^{(r)};$$

- (xi) any retained FBT27-compatible overlap data ρ^{ov} , satisfying the required positivity, Hermiticity, quotient compatibility, and filter-compatibility conditions;

- (xii) a constrained admissible response-data space $\mathfrak{X}_{\text{macro}}^{\text{adm}}$;

- (xiii) an admissible response-metric map $\mathcal{R}_{\text{met}}$.

Then

$$\mathfrak{R}_{\text{adm}}(t) = \mathfrak{P}_{\text{read}}([\mathfrak{S}_t]),$$

and, for every channel,

$$\Pi_{\text{macro}}^{(r)}(t) = \Pi_{\text{res}}^{(r)}(t) \circ \mathbf{P}_{\text{ad}}^{(r)}(t) \circ \Pi_{\text{inv}}^{(r)}(t) \circ \mathbf{P}_{\eta}(t),$$

$$\mu_{\text{macro}}^{(r)}(t) = (\Pi_{\text{macro}}^{(r)}(t))_* \mu_{\text{adm}}(t).$$

For the metric channel,

$$\boxed{\mathfrak{R}_{\text{macro}}(t) = \left(\mathbb{E}_{\mu_{\text{adm}}(t)}[\mathfrak{R}_{\text{adm}}(t) \mid \sigma(\Pi_{\text{macro}}(t))]; \mu_{\text{macro}}(t), \rho^{\text{ov}}(t), \mathcal{C}_{\text{macro}} \right) \in \mathfrak{X}_{\text{macro}}^{\text{adm}}.}$$

The corresponding effective metric is

$$\boxed{g^{\text{eff}}(t) = g^{(0)} + \mathcal{R}_{\text{met}}(\mathfrak{R}_{\text{macro}}(t)).}$$

The complete type-correct chain is

$$\boxed{\begin{aligned} (X_{\text{Reg}}^{(6)}, \Omega_{\mathcal{B}}, H_{\text{str}}) &\longrightarrow \Phi_t \longrightarrow [\mathfrak{S}_t] \xrightarrow{\mathfrak{P}_{\text{read}}} \mathfrak{R}_{\text{adm}}(t), \\ \text{FBT06C/D/E} &\longrightarrow \mu_{\text{adm}}(t), \\ \mathcal{E}_{\text{adm}}(t) &\xrightarrow{\mathbf{P}_{\eta}} \mathcal{E}_{\text{read}}(t) \xrightarrow{\Pi_{\text{inv}}^{(\text{met})}} \mathcal{E}_{\text{inv}}^{(\text{met})}(t), \\ \mathcal{E}_{\text{inv}}^{(\text{met})}(t) &\xrightarrow{\mathbf{P}_{\text{ad}}^{(\text{met})}} \mathcal{E}_{\text{low}}^{(\text{met})}(t) \xrightarrow{\Pi_{\text{res}}^{(\text{met})}} \mathcal{X}_{\text{macro}}^{(\text{met})}(t), \\ \mu_{\text{adm}}(t) &\xrightarrow{(\Pi_{\text{macro}})_*} \mu_{\text{macro}}(t), \\ (\mathfrak{R}_{\text{adm}}(t), \mu_{\text{macro}}(t), \rho^{\text{ov}}(t), \mathcal{C}_{\text{macro}}) &\longrightarrow \mathfrak{R}_{\text{macro}}(t) \xrightarrow{\mathcal{R}_{\text{met}}} \delta g(t), \\ \delta g(t) &\longrightarrow g^{\text{eff}}(t) \xrightarrow{\nabla[g^{\text{eff}}]} R[g^{\text{eff}}]. \end{aligned}}$$

No two consecutive objects are identified. In particular, η constrains first-order transmission, the FBT0C quotient records the microscopic distinctions retained by the metric channel, the FBT08F gap constrains the low-energy survivor sector, Π_{res} specifies the remaining macroscopic resolution, and $\mathcal{R}_{\text{met}}$ controls the final type-changing metric response.

4 Microscopic Mode: Thimble Depth and Inertial Mass

4.1 Local suppression depth and global lift

Let $\mathcal{J}_p$ be an admissible Lefschetz thimble associated with a critical sector p . In a local thimble-adapted description, define the local suppression depth by

$$D_p := \Re \mathcal{A}_{\text{eff}}(p),$$

where $\mathcal{A}_{\text{eff}}$ is the relevant complex action normalization. Globally, the intrinsic depth may be defined only modulo the phase-curvature period lattice:

$$[\mathcal{D}_p] \in \mathbb{R}/\Lambda_{\Omega_{\text{phase}}}.$$

A chosen lift is denoted $\hat{D}_p$.

For an admissible interpolating two-chain Σ_{pq} ,

$$\hat{D}_p - \hat{D}_q = \int_{\Sigma_{pq}} \Omega_{\text{phase}}^{\text{eff}}$$

up to orientation, sign, and period-lattice conventions.

Remark FBT10A-4.1 (Moment level is upstream). The diagonal moment-map level c is neither a mass nor a depth. A safe hierarchy is

$$c \longrightarrow \Omega_{\text{phase}}^{\text{eff}}(c) \longrightarrow [\mathcal{D}_i(c)] \longrightarrow \hat{D}_i(c) \longrightarrow M_i(c).$$

The final arrow is constitutive and is specified below.

4.2 Mass–Depth Constitutive Hypothesis

Hypothesis FBT10A-4.2 (Mass–Depth Constitutive Hypothesis). For an admissible stable sector i , the effective mass coefficient is

$$M_i = M_* Z_i^{\text{pref}} \exp(-\hat{D}_i),$$

where M_* is a reference mass scale and $Z_i^{\text{pref}} > 0$ contains the real physical contribution of intersection multiplicities, fluctuation determinants, coherence factors, branch normalization, and other prefactors.

Consequently,

$$\log \frac{M_i}{M_j} = -(\hat{D}_i - \hat{D}_j) + \log \frac{Z_i^{\text{pref}}}{Z_j^{\text{pref}}}.$$

If the prefactor ratio is slowly varying or subleading, the depth difference gives the leading exponential hierarchy:

$$\log \frac{M_i}{M_j} = -(\hat{D}_i - \hat{D}_j) + O(\log Z^{\text{pref}}).$$

Remark FBT10A-4.3 (Status of the mass law). The mass-depth relation is not derived from the thimble weight alone in FBT10A. It is the leading constitutive identification that connects the semiclassical sector amplitude to the coefficient of the effective worldline action. The operational inertial interpretation follows only after that coefficient appears in the worldline EFT.

4.3 Mass readout versus sector probability

The same upstream depth data may appear in both a mechanical mass law and a semiclassical approximation to a probability density. The resulting quantities nevertheless have different mathematical types.

The mass coefficient is

$$M_i = M_* Z_i^{\text{pref}} e^{-\widehat{D}_i},$$

and has the dimensions of mass. The resolved macroscopic sector probability is

$$p_i(x) = \mu_x(\mathcal{S}_i(x)),$$

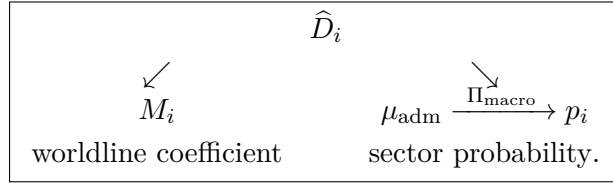
and is dimensionless, with

$$0 \leq p_i \leq 1, \quad \sum_i p_i = 1.$$

Therefore

$$M_i \neq p_i.$$

Principle FBT10A-4.4 (Mass-probability separation). The lifted critical depth $\widehat{D}_i$ may contribute to both the mechanical readout and a saddle approximation to the assembled measure, but these two branches must remain distinct:



Common geometric dependence does not identify the readouts.

Remark FBT10A-4.5 (Metric equivalence need not be mass equivalence). The FBT0C quotient is channel dependent. It is therefore possible that

$$\xi_1 \sim_{\text{met}} \xi_2$$

while

$$M(\xi_1) \neq M(\xi_2).$$

The metric response may be insensitive to a microscopic distinction that remains visible to the thimble-depth mass functional. Conversely, equal mass coefficients need not imply equality of the macroscopic metric response.

A lighter sector is not automatically more or less probable. The probability p_i depends on the complete conditional measure, including the integration domain, within-sector damping, fluctuation-density data, the macroscopic resolution map, and any surviving overlap structure. Thus

$$M_i < M_j$$

implies neither $p_i > p_j$ nor $p_i < p_j$ without additional assumptions.

4.4 Worldline readout and inertial interpretation

Under Principle FBT10A-3.23, the effective action of sector i is

$$S_{\text{eff}}^{(i)}[X, g^{\text{eff}}] = -M_i \int ds_{\text{eff}} + S_{\text{corr}}^{(i)}.$$

If M_i is constant on the region under consideration and the correction terms are neglected, variation gives

$$p_\mu = M_i u_\mu.$$

For an external force satisfying

$$u^\nu \nabla_\nu p_\mu = F_\mu,$$

one obtains

$$F_\mu = M_i a_\mu$$

for constant M_i . Thus the constitutive coefficient has the operational role of inertial mass.

4.5 Conditional Floer representation of relative depth

A compatible Floer trajectory may be represented by

$$u : \mathbb{R} \times S^1 \rightarrow X_{\text{Reg}}^{(6)}$$

or, in a Lagrangian setting,

$$u : \mathbb{R} \times [0, 1] \rightarrow X_{\text{Reg}}^{(6)}.$$

Under the standard compactness, transversality, and normalization conditions,

$$E_{\text{Floer}}(u_{ij}) = \mathcal{A}_H(x_i) - \mathcal{A}_H(x_j)$$

up to sign convention.

Principle FBT10A-4.6 (Floer–Depth Correspondence). When a specified thimble–Floer comparison exists and the action normalization agrees with the curvature-flux convention, one may write

$$\hat{D}_i - \hat{D}_j \sim E_{\text{Floer}}(u_{ij}).$$

Then

$$\frac{M_i}{M_j} \sim \frac{Z_i^{\text{pref}}}{Z_j^{\text{pref}}} \exp[-E_{\text{Floer}}(u_{ij})].$$

The Floer strip or cylinder is an internal symplectic interpolation surface, not the observable spacetime worldline and not a physical string worldsheet. Floer invariance may protect sector organization; it does not imply an Einstein action.

4.6 Balanced shell, the calibrated level, and the number 24

FBT01D relates three logically distinct inputs:

- (i) the non-central Casimir C_T supplied by the gate sector;
- (ii) the positive affine/Berry–Chern level k_S^{aff} supplied by the central sector;
- (iii) the negative central Morse/readout square $\mathbf{S}_{\text{Morse}}^2$.

The balanced branch satisfies

$$\boxed{C_T + \mathbf{S}_{\text{Morse}}^2 = 0, \quad C_T = k_S^{\text{aff}} Q_S^2.}$$

The calibrated rigid result of FBT28B is an independent upstream input:

$$\boxed{k_S^{\text{aff}} = 24.}$$

Hence, on the additional unit-normalized branch $Q_S^2 = 1$,

$$\boxed{C_T = 24.}$$

The logic is therefore

$$\boxed{\underbrace{k_S^{\text{aff}} = 24}_{\text{rigid central calibration}} + \underbrace{C_T = k_S^{\text{aff}} Q_S^2}_{\text{balanced-shell relation}} + \underbrace{Q_S^2 = 1}_{\text{unit branch}} \implies C_T = 24.}$$

Neither C_T nor the balanced shell derives the affine level.

The phase-curvature period lattice depends on the full curvature and cycle data:

$$\boxed{(\Omega_{\text{phase}}, k_S^{\text{aff}}, \text{cycle basis and branch data}) \longrightarrow \Lambda_{\Omega_{\text{phase}}}.$$

Thus a single number 24 does not by itself determine the complete period lattice.

5 Measure-Induced Macroscopic Response

The macroscopic response is built from the probability architecture already established in FBT06D–FBT06E. FBT10A does not introduce a new rule that maps a complex saddle amplitude to a positive probability. Its task is to apply the channel-dependent readout quotient, retain the response observables that survive the adiabatic and resolution filters, and pass the resulting constrained package to $\mathcal{R}_{\text{met}}$.

5.1 Measure-induced sector response

For a resolved diagonal sector decomposition,

$$\boxed{\langle \mathcal{O} \rangle_{\text{macro}}^{\text{diag}}(x) = \sum_{p \in \mathcal{I}_L(x)} p_p(x) \mathcal{O}_p(x), \quad p_p(x) = \mu_x(\mathcal{S}_p(x)).}$$

The probabilities are normalized by construction:

$$\sum_p p_p(x) = 1.$$

A local saddle expansion may provide a semiclassical approximation to the conditional density. Schematically,

$$d\mu_x|_{\mathcal{S}_p} = r_p(\xi) d\lambda_p(\xi), \quad r_p(\xi) \geq 0,$$

with

$$r_p^{\text{sc}}(\xi) = Z_p^{\text{meas}}(\xi) \exp[-\mathcal{D}_p(\xi)]$$

in a leading chart. Therefore

$$p_p(x) = \int_{\mathcal{S}_p(x)} r_p(\xi) d\lambda_p(\xi).$$

The isolated complex saddle coefficient

$$\mathfrak{a}_p = n_p \Delta_p e^{-\widehat{D}_p} \mathcal{C}_p$$

may organize the asymptotic expansion but does not replace this positive measure integral.

Remark FBT10A-5.1 (Analytic provenance). The damping factor is inherited through the chain

$$\begin{array}{lcl} \text{FBT06C :} & \text{absolute convergence} & \\ \Downarrow & & \\ \text{FBT06D :} & \text{fibre probability measure} & \\ \Downarrow & & \\ \text{FBT06E :} & \text{measurable assembly.} & \end{array}$$

FBT10A uses the resulting assembled measure only after convergence and normalization have been established.

5.2 Overlap and coherence corrections

If the FBT27 overlap data remain resolved, the diagonal average is supplemented by

$$\langle \mathcal{O} \rangle_{\text{macro}} = \sum_p p_p \mathcal{O}_{pp} + \sum_{p \neq q} \rho_{pq}^{\text{ov}} \mathcal{O}_{qp}.$$

The diagonal entries are fixed by the probability measure, while the off-diagonal entries describe coherence. In a fully decohered limit, $\rho_{pq}^{\text{ov}} = 0$ for $p \neq q$.

This distinction prevents two opposite errors: overlap data are not reinterpreted as classical probabilities, and probability normalization is not replaced by a transition matrix element.

5.3 Statistical and coherence suppression

Even with positive probabilities, tensor or phase-curvature responses from differently oriented sectors may partially cancel. In a weakly-correlated ensemble one may obtain a model-dependent estimate

$$\left| \sum_p p_p \Omega_{\text{phase},p}^{\text{eff}} \right| \sim \chi_N \left| \Omega_{\text{phase}}^{\text{eff}} \right|_{\text{micro}}, \quad 0 < \chi_N \ll 1.$$

Similarly, off-diagonal coherence norms may be suppressed after coarse-graining. An ideal random-orientation or random-phase model may produce

$$\chi_N \sim N^{-1/2}.$$

This estimate concerns cancellation or decoherence, not probability normalization. It does not determine the metric normalization, Newton's constant, or the low-energy gravitational coupling:

$$\chi_N \neq \mathcal{Z}_{\text{met}} \neq G_{\text{eff}}.$$

A prediction requires a specified ensemble, coherence law, resolution map, dimensional matching, and constitutive normalization.

5.4 Complementarity with FBT12A

FBT12A supplies a six-dimensional curvature–tension response equation. FBT10A supplies the measure-induced coarse-graining and constitutive architecture by which admissible response data can become a symmetric metric deformation. Neither is a direct theorem consequence of the other. They meet only after the response equation has been reduced, its real zero modes and branch data have been identified, the FBT06 measure architecture has been assembled, and the constitutive metric map has been supplied.

6 Explicit Local Representative: A Non-Flat T^2 Connection

The purpose of this section is to illustrate the tensor-type conversion within the fixed FBT signature convention $(-+++)$. It is not a derivation of the constitutive law or of Einstein dynamics.

6.1 Local bundle geometry

Let the base be Minkowski space

$$\mathcal{M}_4 = \mathbb{R}^{1,3}, \quad x^\mu = (t, x, y, z),$$

with

$$\boxed{\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1).}$$

Consider the locally trivial principal bundle

$$T^2 \hookrightarrow X = \mathcal{M}_4 \times T^2 \xrightarrow{\pi} \mathcal{M}_4.$$

Let (ψ^1, ψ^2) be the fibre angles and $\{\tau_1, \tau_2\}$ a basis of $\mathfrak{t}^2$. Define

$$A = A^1 \tau_1 + A^2 \tau_2,$$

with

$$A^1 = d\psi^1 + \frac{B}{2}(x dy - y dx),$$

and

$$A^2 = d\psi^2 + \frac{E}{2}(t dz - z dt).$$

6.2 Connection curvature

Since T^2 is Abelian,

$$F_A = dA = B dx \wedge dy \tau_1 + E dt \wedge dz \tau_2.$$

Thus local triviality does not imply flatness.

6.3 Quadratic response tensor

Choose

$$\langle \tau_a, \tau_b \rangle_V = \delta_{ab}$$

and

$$H^{\alpha\beta} = \eta^{\alpha\beta} = \text{diag}(-1, 1, 1, 1).$$

Define

$$Q_{\mu\nu} := \sum_{a=1}^2 F_{\mu\alpha}^a F_{\nu\beta}^a \eta^{\alpha\beta}.$$

The nonzero components are

$$F_{xy}^1 = B, \quad F_{tz}^2 = E,$$

with their antisymmetric partners. Direct calculation gives

$$\boxed{Q_{\mu\nu} = \text{diag}(E^2, B^2, B^2, -E^2).}$$

The tensor is symmetric even though the curvature is antisymmetric.

6.4 Perturbative response metric

Let $\varepsilon \in \mathbb{R}$ be a local constitutive coefficient and define

$$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + \varepsilon Q_{\mu\nu}.$$

Then

$$g_{\mu\nu}^{\text{eff}} = \text{diag} \left(-1 + \varepsilon E^2, 1 + \varepsilon B^2, 1 + \varepsilon B^2, 1 - \varepsilon E^2 \right).$$

Its determinant is

$$\det g^{\text{eff}} = -(1 - \varepsilon E^2)^2 (1 + \varepsilon B^2)^2.$$

A convenient sufficient condition for signature $(-+++)$ is

$$\varepsilon E^2 < 1, \quad 1 + \varepsilon B^2 > 0.$$

For $\varepsilon > 0$, the second condition is automatic and the first becomes $0 \leq \varepsilon E^2 < 1$.

The background trace is

$$\sigma_Q := \eta^{\mu\nu} Q_{\mu\nu} = 2(B^2 - E^2).$$

Hence

$$Q_{\mu\nu} = \frac{1}{4} \sigma_Q \eta_{\mu\nu} + \Sigma_{\mu\nu}^{(Q)}, \quad \eta^{\mu\nu} \Sigma_{\mu\nu}^{(Q)} = 0.$$

Remark FBT10A-6.1 (Scope of the local example). The example proves only that a non-flat relative-phase connection, together with a vertical pairing and a symmetric contraction tensor, can produce a real symmetric response tensor and a controlled Lorentzian metric deformation. It does not determine ε , select the response uniquely, produce a conserved source tensor, or prove the Einstein equation.

7 Central Zero-Mode Interface and Conditional Palatini Closure

7.1 Division of roles

The updated hierarchy is:

FBT10A :	general response-metric architecture,
FBT12A :	six-dimensional curvature–tension response equation,
FBT12C :	central zero-mode equation and response tuple $\mathfrak{R}_S$,
FBT12D :	central specialization and conditional Palatini completion.

The meeting relation is

$$\mathcal{C}_{\text{met}} = \mathcal{R}_{\text{met}} \circ \iota_S.$$

FBT12C may supply an Abelian equation of the schematic form

$$\nabla^\mu F_{\mu\nu}^S = J_\nu^S + O(E^2/M_{\text{KK}}^2),$$

but

$$F^S \neq R^{ab}, \quad F^S \neq g^{\text{eff}}.$$

The central two-form becomes metric input only through the constitutive specialization.

7.2 Off-shell coframe and Palatini action

Once g^{eff} has been promoted to an independent low-energy field, a local coframe class $[e]$ satisfies

$$g_{\mu\nu}^{\text{eff}} = \eta_{ab} e^a{}_\mu e^b{}_\nu, \quad \eta_{ab} = \text{diag}(-1, 1, 1, 1).$$

A representative coframe e^a and an independent Lorentz connection ω^{ab} may be used in the first-order action

$$S_{\text{Palatini}} = \frac{1}{16\pi G_{\text{eff}}} \int \epsilon_{abcd} \left(e^a \wedge e^b \wedge R^{cd}[\omega] - \frac{\Lambda_{\text{eff}}}{6} e^a \wedge e^b \wedge e^c \wedge e^d \right) + S_{\text{matter}}.$$

The precise normalization of the cosmological term follows the stated convention and may be absorbed into the definition of Λ_{eff} .

Under coframe nondegeneracy, locality, Palatini dominance, and a consistent matter coupling, the field equations take an Einstein-type form in the torsion-suppressed branch:

$$G_{\mu\nu}^{\text{eff}} + \Lambda_{\text{eff}} g_{\mu\nu}^{\text{eff}} = 8\pi G_{\text{eff}} T_{\mu\nu}^{\text{eff}} + O(E^2/M_{\text{KK}}^2).$$

7.3 Spin current and torsion boundary

FBT08G supplies a classical worldline spin tensor on a persistent massive branch. In a first-order gravitational theory, a nonzero spin current can source torsion. Therefore the torsionless Einstein-like closure is not universal.

Principle FBT10A-7.1 (Spin–torsion interface). The torsion-suppressed Palatini branch applies when matter spin currents are absent, integrated out, incoherently averaged, or parametrically negligible at the scale of the closure. If the FBT08G spin sector is resolved and couples to the independent Lorentz connection, the natural extension may be Einstein–Cartan type rather than strictly torsionless Einstein gravity.

The existence of this possible extension does not alter the response–metric construction; it affects the downstream variational completion and the connection field equation.

8 Structural Consequences and Boundaries

The revised architecture supports the following controlled statements.

1. **Hamiltonian and thimble flows are distinct.** The physical carrier state is transported by $X_{H_{\text{str}}}$, whereas integration-cycle selection is controlled by the complex-Morse flow. Compatibility is an assumption, not an identity.
2. **Hamiltonian invariant conservation does not imply readout reconstructibility.** Although

$$\Phi_t^*(\Omega_B^{\wedge k}) = \Omega_B^{\wedge k},$$

a response map depending on a higher-order invariant may be many-to-one. The microscopic evolution can remain reversible while the macroscopic readout loses the ability to distinguish lower-order representatives.

3. **Mass ratios have a leading depth hierarchy.**

$$\log \frac{M_i}{M_j} = -(\hat{D}_i - \hat{D}_j) + \log \frac{Z_i^{\text{pref}}}{Z_j^{\text{pref}}}.$$

The pure exponential formula is the leading approximation when the prefactor ratio is subleading.

4. **Floer theory is a conditional microscopic interface.** A compatible Floer action difference may represent a relative depth, but Floer homology does not determine the macroscopic derivative expansion or the Einstein action.
5. **The balanced shell does not derive the affine level.** The calibrated rigid input $k_S^{\text{aff}} = 24$, the balanced relation $C_T = k_S^{\text{aff}} Q_S^2$, and the unit condition $Q_S^2 = 1$ together imply $C_T = 24$.
6. **The relative-phase vertical sector is isotropic.** The mother symplectic form has $\Omega_B = \omega_H + \eta$ on the regular branch. Phase curvature and connection curvature remain separate data.
7. **The mixed form determines first-order readability.** The map

$$\eta_V^b : V \rightarrow H_A^*$$

identifies the vertical directions that can affect horizontal readout at first order. On the Liouville-saturated rank-two branch it has full rank, so η supplies a complete mixed transmission channel rather than eliminating vertical directions.

8. **First-order readability is not invariant-order resolution.** A channel readout $\mathcal{Q}^{(r)}$ may identify distinct readable states. The quotient

$$\Pi_{\text{inv}}^{(r)} : \mathcal{E}_{\text{read}} \rightarrow \mathcal{E}_{\text{read}} / \sim_r$$

records precisely which distinctions survive in channel r .

9. **Quadratic metric response is generically sign blind.** For the curvature-quadratic channel,

$$Q[-\mathcal{K}] = Q[\mathcal{K}].$$

Thus the metric response may be unable to reconstruct the sign of a microscopic curvature representative even though the linear curvature channel can distinguish it.

10. **Readout equivalence is channel dependent.** Metric equivalence does not imply mass equivalence, probability equivalence, or overlap equivalence. Distinct effective sectors can retain different quotients of one microscopic state space.
11. **Invariant-order resolution is not persistence.** A readable invariant class may still lie in the gapped transverse spectrum. The FBT08F selector $P_{\text{ad}}^{(r)}$ retains the collective or resolved-soft sector and suppresses discarded modes up to adiabatic corrections.
12. **Persistence is not macroscopic resolution.** The residual map $\Pi_{\text{res}}^{(r)}$ determines which low-energy invariant classes remain distinguished by the final macroscopic description.
13. **The macroscopic map is channel dependent and factorised.**

$$\boxed{\Pi_{\text{macro}}^{(r)} = \Pi_{\text{res}}^{(r)} \circ P_{\text{ad}}^{(r)} \circ \Pi_{\text{inv}}^{(r)} \circ P_{\eta}.$$

Thus macroscopic indistinguishability may have structural, dynamical, and resolution-induced origins.

14. **Probability normalisation is inherited upstream.** FBT06D supplies the normalised fibre measures ν_b , and FBT06E assembles them into μ_{adm} . FBT10A does not introduce a second amplitude-to-probability rule.

15. **Macroscopic probabilities are induced by the selected channel pushforward.**

For the metric channel,

$$\mu_{\text{macro}} = (\Pi_{\text{res}}) * (\text{P}_{\text{ad}}) * (\Pi_{\text{inv}}) * (\text{P}_{\eta}) * \mu_{\text{adm}}, \quad p_i = \mu_x(\mathcal{S}_i).$$

16. **Mass and probability remain distinct.** The mass M_i is a dimensionful worldline coefficient, whereas p_i is a normalised dimensionless probability. Common depth dependence does not imply $M_i = p_i$.

17. **Probability and overlap remain distinct.** The diagonal entries p_i are inherited from the FBT06 measure architecture. Surviving off-diagonal data ρ_{ij}^{ov} belong to an FBT27-compatible overlap/coherence construction and must survive the selected invariant quotient, adiabatic filter, and residual resolution.

18. **The response-data space is constrained.** The entries $A, F_A, \eta, g^{\text{read}}, F^{\text{read}}$ obey compatibility relations and must not be treated as independent duplicate sources.

19. **The metric is a constitutive response variable.** The symplectic form or wedge degree alone does not determine it. The explicit response law is model dependent, while the layered architecture is the proposed universal structure.

20. **A particle source and a metric response require matching.** The stress tensor and the constitutive perturbation of response data are candidate descriptions of related macroscopic information. They must be derived from a common action or explicitly matched.

21. **The variational source uses an off-shell metric.** The constitutive relation is imposed only after variation or through a separate matching equation. If the mass depends on the metric, the additional chain-rule source terms must be included.

22. **Einstein gravity is a conditional infrared closure.** It requires an independent variational completion, locality, diffeomorphism covariance, derivative ordering, and a nonzero Ricci-scalar coefficient.

23. **Resolved spin may require torsion.** The torsionless closure is limited to a spinless or spin-current-suppressed branch. A resolved FBT08G spin sector may require an Einstein-Cartan extension.

24. **Readout is not observation.** FBT10A constructs mass and metric readout variables, not experimental outcomes.

8.1 The ten-component symmetric response interface

A symmetric rank-two tensor in four dimensions decomposes as

$$10_{\text{sym}} = 1_{\text{trace}} + 9_{\text{traceless}}.$$

After an Einstein-type closure, these components naturally interface with the scalar and traceless-Ricci sectors. This is a tensorial matching statement, not a pre-closure identification with the Ricci tensor. The Weyl sector remains controlled by the gravitational field equations, Bianchi identities, and boundary data.

9 Conclusion

FBT10A v2.6 identifies a response-metric bridge compatible with the current FBT0B, FBT0C, FBT01A, FBT01D, FBT06C–FBT06E, FBT08F, FBT08G, and FBT27 architectures. The global carrier organisation is

$$T_{\text{rel}}^2 \hookrightarrow X_{\text{Reg}}^{(6)} \xrightarrow{\pi_{\text{fib}}} B_{\text{Reg}}^{(4)},$$

and the mother symplectic form on the regular isotropic-fibre sector is

$$\Omega_{\mathcal{B}} = \omega_H + \eta.$$

No second six-dimensional-to-four-dimensional projection is introduced. The bundle projection is separated from tensorial readout descent, measure assembly, first-order mixed readability, invariant-order quotienting, adiabatic selection, and macroscopic resolution.

The physical Hamiltonian flow and the complex-Morse thimble flow remain distinct:

$$\boxed{X_{H_{\text{str}}} \neq -\text{grad } \Re \mathcal{S}_{\mathbb{C}}.}$$

The first transports the structural state; the second selects the semiclassical integration sectors. Their compatibility defines the admissible branch on which the mechanical, probabilistic, and metric readouts are constructed.

The microscopic mechanical branch is governed by

$$\boxed{M_i = M_* Z_i^{\text{pref}} e^{-\widehat{D}_i}.$$

Depth differences therefore give the leading exponential mass hierarchy. The combined FBT08E, FBT15C, and FBT08F interface supplies the collective worldline, the positive transverse physical stability operator, and the dynamical adiabatic condition

$$\varepsilon_{\text{ad}} = \frac{\Omega_{\text{ext}}}{\omega_{\text{gap}}} \ll 1.$$

With the $(- + ++)$ convention, the leading worldline term is

$$S_{\text{worldline}}^{(i)} = -M_i \int \sqrt{-g_{\mu\nu}^{\text{eff}} dx^\mu dx^\nu} + S_{\text{corr}}^{(i)}.$$

The probabilistic branch is inherited rather than postulated:

$$\boxed{\text{FBT06C} \longrightarrow \text{FBT06D} \longrightarrow \text{FBT06E},}$$

corresponding to absolute convergence, fibrewise probability normalisation, and measurable assembly over the regular relative-phase bundle.

The first response layer is the FBT01A mixed-readability map

$$\eta_V^{\flat} : V \rightarrow H_A^*.$$

On the regular Liouville-saturated rank-two branch, $\text{rank } \eta_V^{\flat} = 2$, so every vertical direction can enter the first-order horizontal transmission channel. This stage is not a lossy coarse-graining.

FBT0C v1.5 supplies the missing structural quotient. For each response channel r ,

$$\xi_1 \sim_r \xi_2 \iff \mathcal{Q}^{(r)}(\xi_1) = \mathcal{Q}^{(r)}(\xi_2)$$

defines the quotient

$$\Pi_{\text{inv}}^{(r)} : \mathcal{E}_{\text{read}} \rightarrow \mathcal{E}_{\text{inv}}^{(r)}.$$

Whenever $\mathcal{Q}^{(r)}$ is non-injective, the channel loses the ability to reconstruct every lower-order microscopic representative. This loss of distinguishability is compatible with exact Hamiltonian preservation of the symplectic wedge invariants:

$$\boxed{\text{invariant conservation} \neq \text{microscopic reconstructibility from the invariant.}}$$

The underlying Hamiltonian dynamics need not become irreversible.

FBT08F then supplies the adiabatic low-energy selector. Assuming the spectral projection descends to the channel quotient, gapped transverse modes are integrated out up to derivative-suppressed corrections. Only after first-order readability, invariant-order resolution, and dynamical selection is the residual macroscopic resolution applied. Thus

$$\boxed{\Pi_{\text{macro}}^{(r)} = \Pi_{\text{res}}^{(r)} \circ \mathbf{P}_{\text{ad}}^{(r)} \circ \Pi_{\text{inv}}^{(r)} \circ \mathbf{P}_{\eta}.$$

This establishes the four-level distinction

$$\boxed{\begin{aligned} &\text{readability} \neq \text{invariant-order distinguishability,} \\ &\text{invariant-order distinguishability} \neq \text{dynamical persistence,} \\ &\text{dynamical persistence} \neq \text{macroscopic resolution.} \end{aligned}}$$

For the metric channel, the macroscopic measure is

$$\boxed{\mu_{\text{macro}} = (\Pi_{\text{res}})_* (\mathbf{P}_{\text{ad}})_* (\Pi_{\text{inv}})_* (\mathbf{P}_{\eta})_* \mu_{\text{adm}}.}$$

Macroscopic response variables are conditional expectations with respect to this measure. In a sector-diagonal regime,

$$\langle \mathcal{O} \rangle_{\text{macro}} = \sum_i p_i \mathcal{O}_i, \quad p_i = \mu_x(\mathcal{S}_i), \quad \sum_i p_i = 1.$$

The mass coefficient and the probability have different types and roles:

$$\boxed{M_i \neq p_i.}$$

If coherence between distinct admissible sectors survives the metric quotient, the adiabatic selector, and the final resolution, an FBT27-compatible overlap sector supplements the diagonal response.

The new quotient layer also sharpens the meaning of the title. Mass and gravity are dual readout sectors because they can impose different equivalence relations on the same microscopic state space. It is possible that

$$\xi_1 \sim_{\text{met}} \xi_2$$

while

$$M(\xi_1) \neq M(\xi_2).$$

Conversely, equal masses need not imply equal metric responses. Common microscopic origin does not require common readout resolution.

The curvature-quadratic response provides a concrete example:

$$Q_{\mu\nu}[-\mathcal{K}] = Q_{\mu\nu}[\mathcal{K}].$$

Therefore a metric contribution depending only on Q is insensitive to at least the sign of the microscopic curvature representative. This is a type-changing, invariant-induced readout quotient, not an arbitrary statistical deletion of dimensions.

The macroscopic metric branch is consequently

$$\begin{aligned}
[\mathfrak{S}_t] &\xrightarrow{\mathfrak{P}_{\text{read}}} \mathfrak{R}_{\text{adm}}(t), \\
\mu_\eta(t) &:= (\mathbf{P}_\eta)_* \mu_{\text{adm}}(t), \\
\mu_{\text{inv}}(t) &:= (\Pi_{\text{inv}})_* \mu_\eta(t), \\
\mu_{\text{low}}(t) &:= (\mathbf{P}_{\text{ad}})_* \mu_{\text{inv}}(t), \\
\mu_{\text{macro}}(t) &:= (\Pi_{\text{res}})_* \mu_{\text{low}}(t), \\
(\mathfrak{R}_{\text{adm}}, \mu_{\text{macro}}, \rho^{\text{ov}}, \mathcal{C}_{\text{macro}}) &\longrightarrow \mathfrak{R}_{\text{macro}}(t) \xrightarrow{\mathcal{R}_{\text{met}}} \delta g(t), \\
\delta g(t) &\longrightarrow g^{\text{eff}}(t) = g^{(0)} + \delta g(t).
\end{aligned}$$

The response package is real, gauge covariant, normalised, and constrained. The metric is not a direct projection of a curvature two-form or of an exterior degree; it is the output of a type-changing constitutive law acting on channel-quotiented macroscopic data.

The factorisation sharply reduces the undifferentiated constitutive freedom previously assigned to Π_{macro} . First-order transmission is constrained by FBT01A, the invariant quotient by FBT0C once the channel readout is fixed, and low-energy persistence by FBT08F. The genuinely residual choices are the final channel resolution and the independent response-metric law $\mathcal{R}_{\text{met}}$.

The central level remains a positive independent input. On the rigid calibrated branch,

$$k_S^{\text{aff}} = 24.$$

FBT01D then gives

$$C_T = k_S^{\text{aff}} Q_S^2,$$

and the unit branch $Q_S^2 = 1$ yields $C_T = 24$. The logical arrow is not $C_T \rightarrow k_S^{\text{aff}}$.

A stable particle configuration may be represented as a variational source and as a perturbation of the constitutive response data, the assembled measure, and possibly the channel maps. These are not independent duplicate sources. Their consistency must be established by a common effective action or an explicit matching condition. For the source variation, g^{eff} is treated off shell as an independent low-energy variable.

Finally, the central specialisation

$$\mathcal{C}_{\text{met}} = \mathcal{R}_{\text{met}} \circ \iota_S$$

feeds the FBT12D variational completion. The Palatini action includes the cosmological term and gives a torsion-suppressed Einstein-type closure only in the appropriate branch. A resolved spin current from FBT08G may require an Einstein–Cartan extension.

The complete division of responsibilities is

FBT06C :	absolute convergence,
FBT06D :	fibrewise probability normalisation,
FBT06E :	measurable probability assembly,
FBT01A :	first-order mixed readability,
FBT0C :	invariant-order readout quotients,
FBT08F :	adiabatic low-energy persistence,
FBT27 :	inter-sector overlap and coherence,
FBT10A :	residual resolution and response-metric construction.

Mass and gravity remain dual readout sectors, while probability, readability, invariant-order distinguishability, and adiabatic persistence provide the layered bridge from the convergent microscopic theory to the macroscopic response. Their unity is a controlled hierarchy, not an identity of curvature forms, masses, probabilities, overlaps, invariants, or projection maps.

A Effective-Field-Theory Boundary of an Einstein-Type Closure

The leading Einstein-type metric closure requires logically independent inputs:

- (i) an already selected off-shell Lorentzian metric field;
- (ii) four-dimensional locality;
- (iii) diffeomorphism invariance;
- (iv) single-metric dominance;
- (v) a derivative expansion;
- (vi) suppression of higher-curvature and nonlocal terms;
- (vii) a nonzero coefficient of the Ricci scalar;
- (viii) a variational matching between the constitutive branch and the low-energy field variables.

At zeroth derivative order, a cosmological term is allowed. There is no nontrivial generally covariant scalar of a single metric at first derivative order. At two derivatives, the Ricci scalar gives the leading non-boundary term:

$$S_{\text{metric}} = \int d^4x \sqrt{-g} \left[-\Lambda_0 + \frac{M_{\text{Pl,eff}}^2}{2} R + \dots \right].$$

The derivative ordering does not determine the sign or magnitude of $M_{\text{Pl,eff}}^2$, Λ_0 , or the matching coefficients. Those require input from the response sector.

If resolved spin currents couple to an independent Lorentz connection, the low-energy completion may contain torsion and additional Einstein–Cartan operators. The pure metric expansion above then represents the torsion-suppressed or torsion-integrated-out branch.

B Response-Tensor Decomposition and the Ricci Interface

Let

$$\mathcal{S}_{\mu\nu}$$

be a real symmetric macroscopic response tensor on $(B_{\text{Reg}}^{(4)}, g^{\text{eff}})$. Its trace decomposition is

$$\boxed{\mathcal{S}_{\mu\nu} = \frac{1}{4}\sigma g_{\mu\nu}^{\text{eff}} + \Sigma_{\mu\nu}},$$

where

$$\sigma := (g^{\text{eff}})^{\mu\nu} \mathcal{S}_{\mu\nu}, \quad (g^{\text{eff}})^{\mu\nu} \Sigma_{\mu\nu} = 0.$$

Thus

$$\text{Sym}^2(T^*B_{\text{Reg}}^{(4)}) = \mathbb{R} g^{\text{eff}} \oplus \text{Sym}_0^2(T^*B_{\text{Reg}}^{(4)}),$$

with

$$10_{\text{sym}} = 1_{\text{trace}} + 9_{\text{traceless}}.$$

This matches the algebraic decomposition of the Ricci sector only after a gravitational closure has been imposed. Suppose, conditionally,

$$G_{\mu\nu}[g^{\text{eff}}] + \Lambda_{\text{eff}} g_{\mu\nu}^{\text{eff}} = 8\pi G_{\text{eff}} \mathcal{S}_{\mu\nu}.$$

Taking the trace gives

$$-R[g^{\text{eff}}] + 4\Lambda_{\text{eff}} = 8\pi G_{\text{eff}} \sigma,$$

and the traceless part gives

$$R_{\mu\nu} - \frac{1}{4}R g_{\mu\nu}^{\text{eff}} = 8\pi G_{\text{eff}} \Sigma_{\mu\nu}.$$

Hence the trace and traceless response sectors interface with the scalar and traceless-Ricci sectors. This is conditional on the Einstein-type closure; it is not an identity between a pre-metric response tensor and the Ricci tensor.

The Weyl tensor has a different tensor type and is not locally fixed by $\mathcal{S}_{\mu\nu}$ alone. Its propagation depends on the field equations, Bianchi identities, and boundary data.

C Relation to FBT01A and the Isotropic Relative-Phase Fibre

FBT01A distinguishes the mother symplectic form, its mixed component, the relative-phase connection, the connection curvature, and the phase curvature:

$$\Omega_{\mathcal{B}}, \quad \eta, \quad A, \quad F_A, \quad \Omega_{\text{phase}}^{\text{eff}}.$$

On the regular isotropic-fibre sector,

$$\Omega_{\mathcal{B}} = \omega_H + \eta, \quad \Omega_{\mathcal{B}}|_{V \times V} = 0.$$

The Liouville identity is

$$\frac{\Omega_{\mathcal{B}}^{\wedge 3}}{3!} = \frac{\omega_H \wedge \eta^{\wedge 2}}{2}.$$

A local mixed density may be defined by

$$\frac{1}{2} \eta \wedge \eta \wedge \omega_H = \rho_{\text{mix}} \frac{1}{3!} \Omega_{\mathcal{B}}^{\wedge 3},$$

but the exact normalized ratio on the regular saturated branch is not an independent coupling constant. The mixed sector is one structural input to FBT10A, not the complete gravitational source.

The division of roles is

FBT01A :	local carrier pairing and mixed readout geometry,
FBT10A :	macroscopic constitutive response metric,
FBT12C :	central zero-mode response tuple,
FBT12D :	central metric specialization and variational closure.

D Logical Status of the Main Inputs

Statement	Required input	Status
Global T_{rel}^2 -fibration	FBT0B principal-bundle architecture	Upstream input
Isotropic vertical mother-form sector	FBT01A mixed structure	Upstream input
Hamiltonian structural flow	FBT0F and an admissible H_{str}	Upstream input
Lefschetz–Morse sector selection	Complex action, metric, and thimble hypotheses	Independent input
Compatibility of the two flows	No destructive wall crossing or loss of admissibility	Assumption
Reduced collective kernel	FBT15C/FBT08F Morse–Bott reduction	Upstream input
Positive transverse stability operator	Reduced coercivity	Conditional theorem upstream
Dynamical gap ω_{gap}	Kinetic operator and positive transverse spectrum	Derived from input
Mass-depth relation	$M_i = M_* Z_i^{\text{pref}} e^{-\widehat{D}_i}$	Constitutive hypothesis
Fibre probability measures ν_b	FBT06D damping measure and normalization	Upstream theorem
Assembled admissible measure μ_{adm}	FBT06E measurable probability kernel and disintegration	Upstream assembly input
First-order mixed-readability map $\mathbf{P}_\eta$	FBT01A mixed form η and its vertical map $\eta_V^\flat$	Geometrically constrained transmission
Invariant-order quotient $\Pi_{\text{inv}}^{(r)}$	FBT0C channel readout $\mathcal{Q}^{(r)}$ and equivalence relation	Derived readout quotient
Adiabatic low-energy selector $\mathbf{P}_{\text{ad}}^{(r)}$	FBT08F spectral projection plus quotient–adiabatic compatibility	Conditional dynamical reduction
Residual resolution map $\Pi_{\text{res}}^{(r)}$	Choice of final macroscopic resolution and measurable partition	Constitutive input
Channel coarse-graining map $\Pi_{\text{macro}}^{(r)}$	$\Pi_{\text{macro}}^{(r)} = \Pi_{\text{res}}^{(r)} \circ \mathbf{P}_{\text{ad}}^{(r)} \circ \Pi_{\text{inv}}^{(r)} \circ \mathbf{P}_\eta$	Derived composite map
Sector probabilities p_i	Conditional measures $p_i = \mu_x(\mathcal{S}_i)$	Derived from μ_{adm} and the selected channel map Π_{macro}
Off-diagonal overlap data ρ_{ij}^{ov}	FBT27-compatible overlap and coherence construction	Optional upstream/constitutive interface
Response-metric map $\mathcal{R}_{\text{met}}$	Constrained real response data and calibration	Constitutive input

Continued on next page

Statement	Required input	Status
Global response metric	Equivariant local maps and signature admissibility	Conditional descent theorem
$k_S^{\text{aff}} = 24$	FBT28B rigid central calibration	Upstream calibrated input
$C_T = 24$	$k_S^{\text{aff}} = 24$, balanced shell, and $Q_S^2 = 1$	Conditional consequence
Point-particle source formula	Off-shell metric and fixed-branch mass convention	Conditional EFT result
Einstein-type closure	Variational completion and IR assumptions	Conditional downstream closure
Torsionless closure with spin	Absent, averaged, or integrated-out spin current	Restricted branch

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